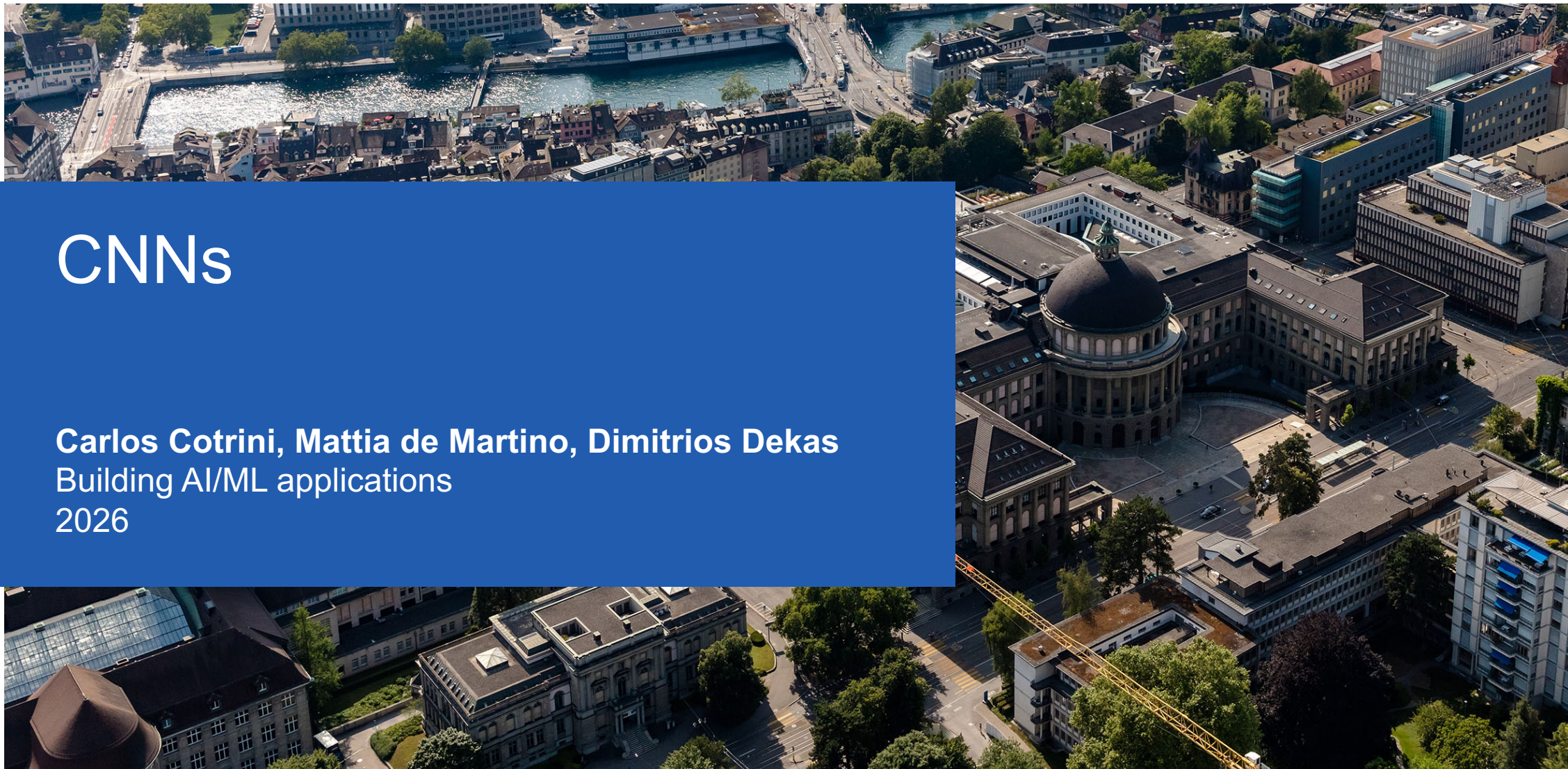


CNNs

Carlos Cotrini, Mattia de Martino, Dimitrios Dekas
Building AI/ML applications
2026



Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

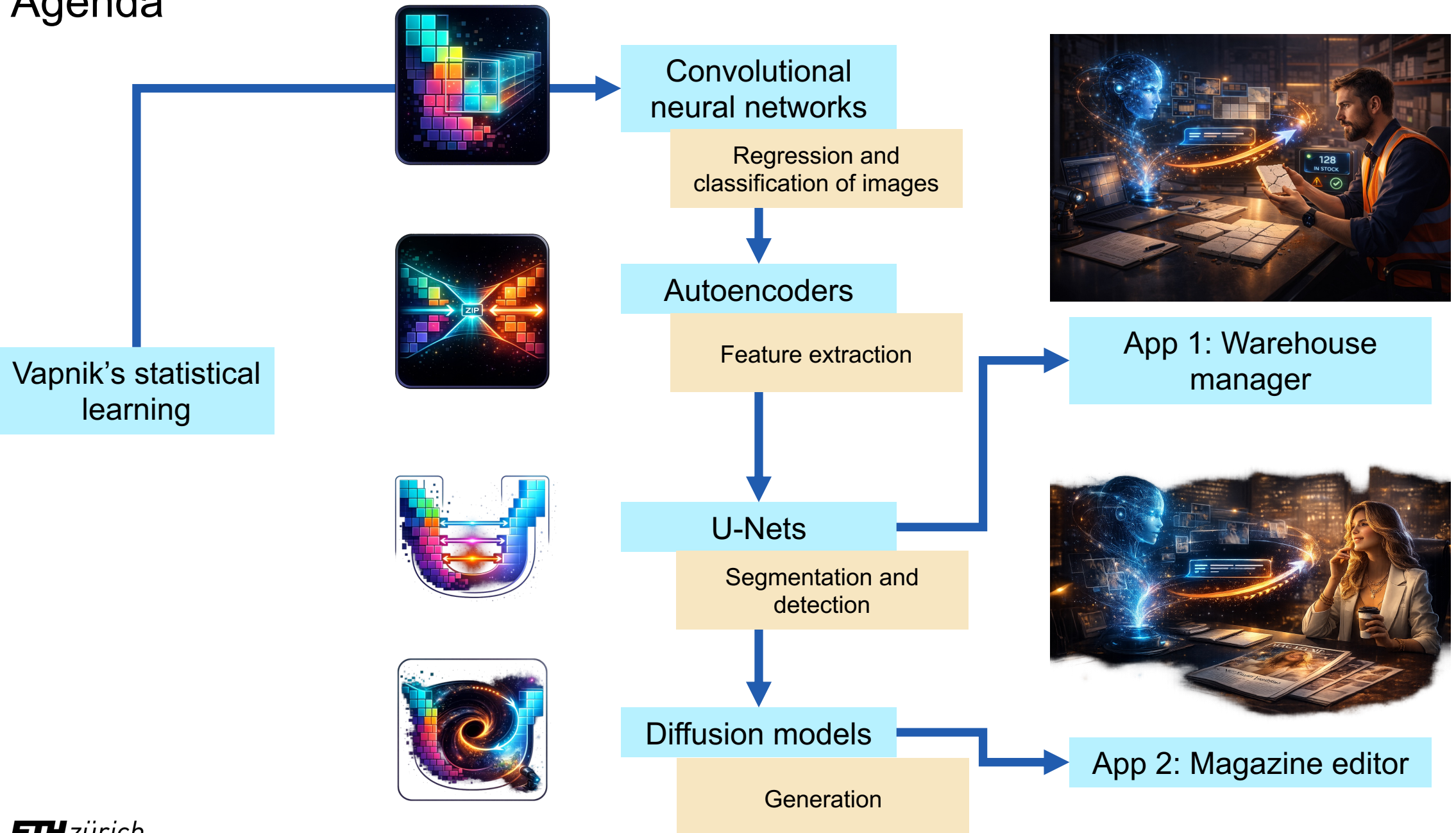
Segmentation and
detection



Generation



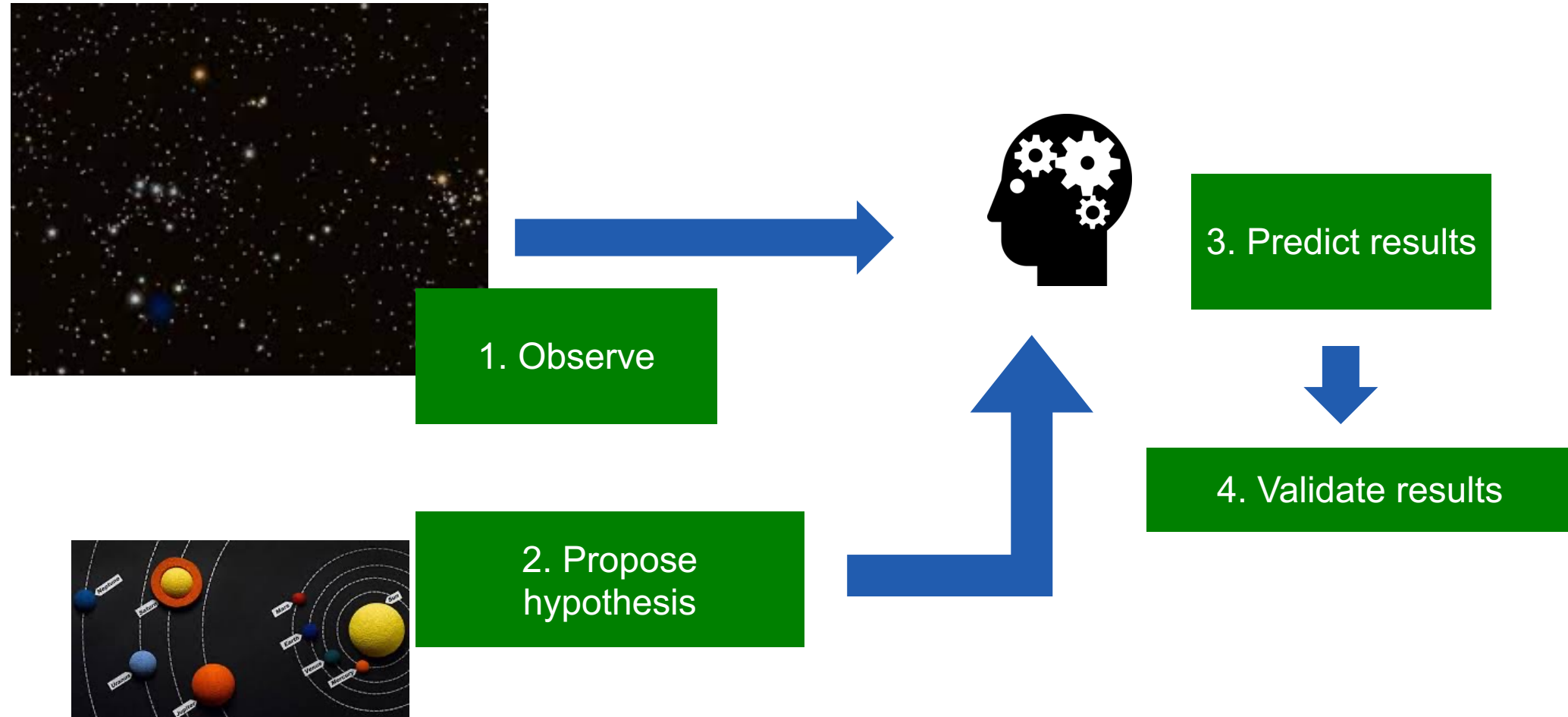
Agenda



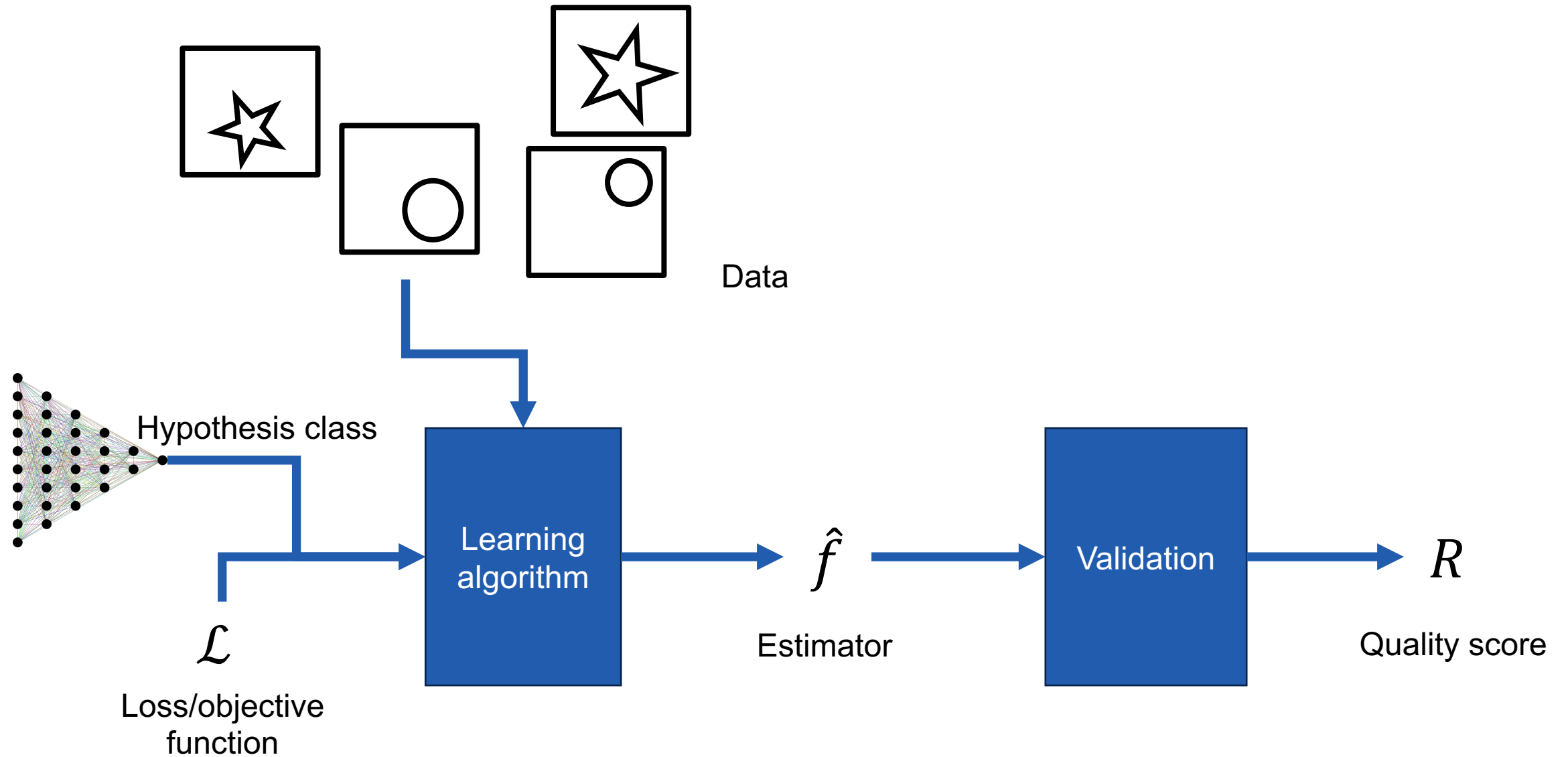
Motivation

What is learning?

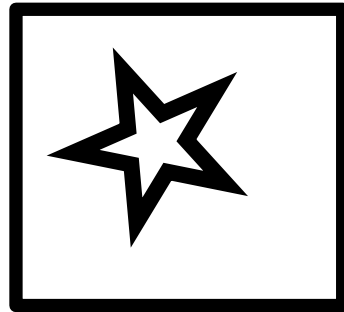
How humans create knowledge



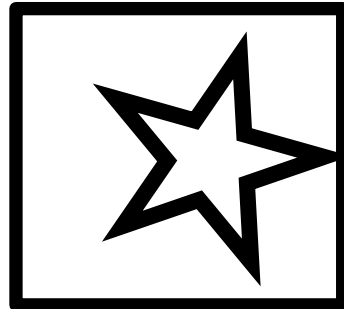
Statistical learning in data science



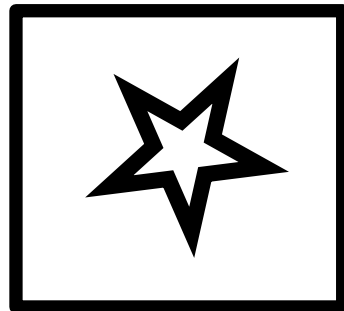
Classifying stars and planets



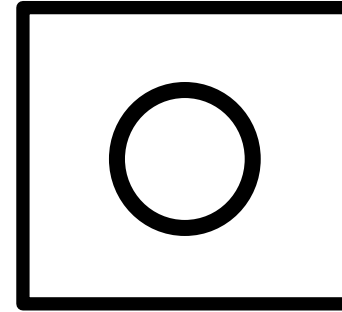
Star



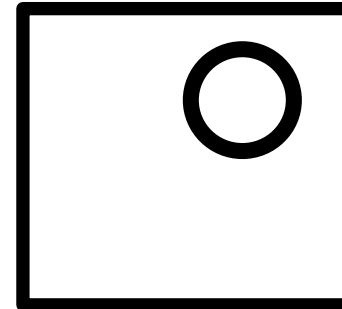
Star



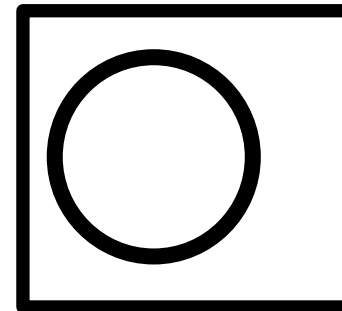
Star



Planet

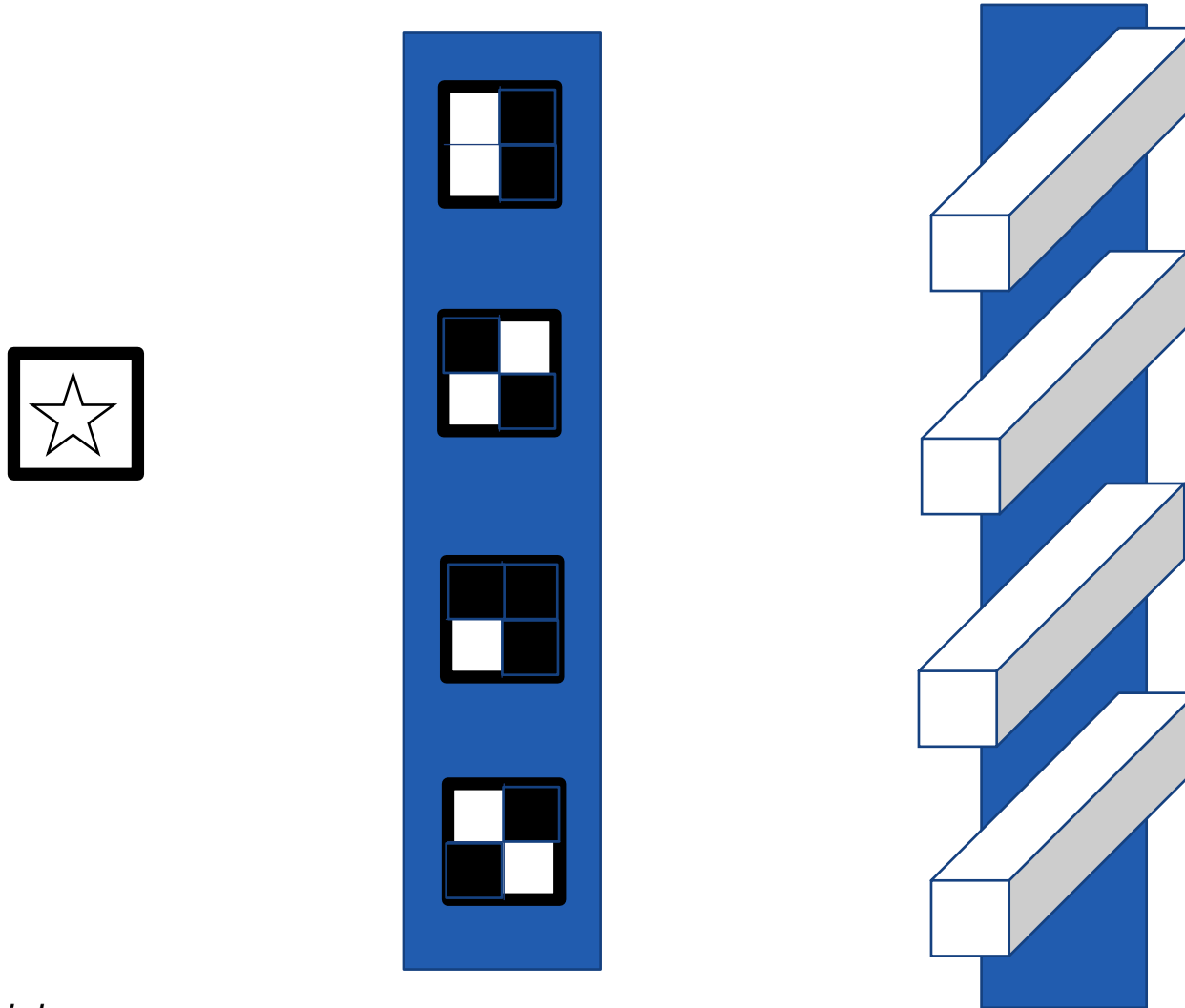


Planet

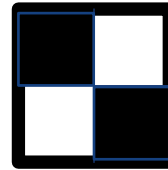
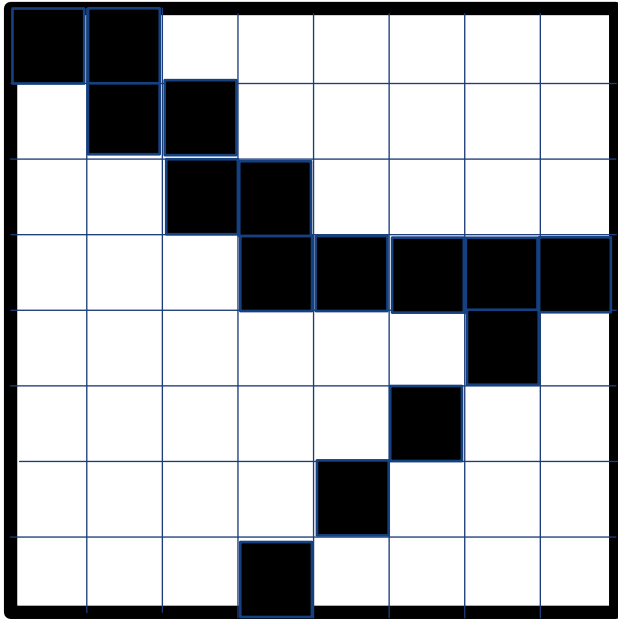


Planet

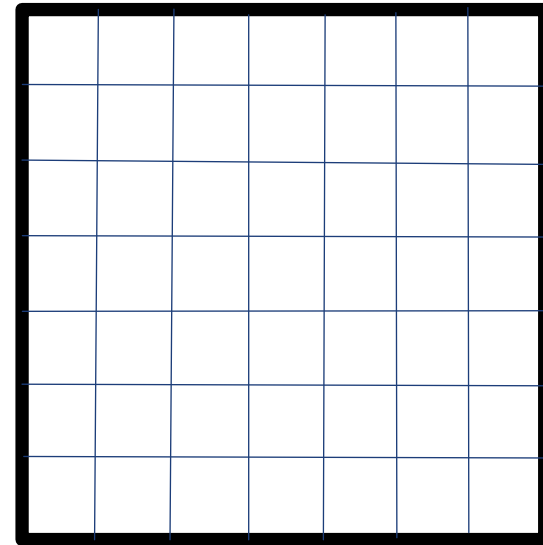
Convolutional neural networks



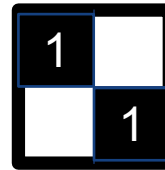
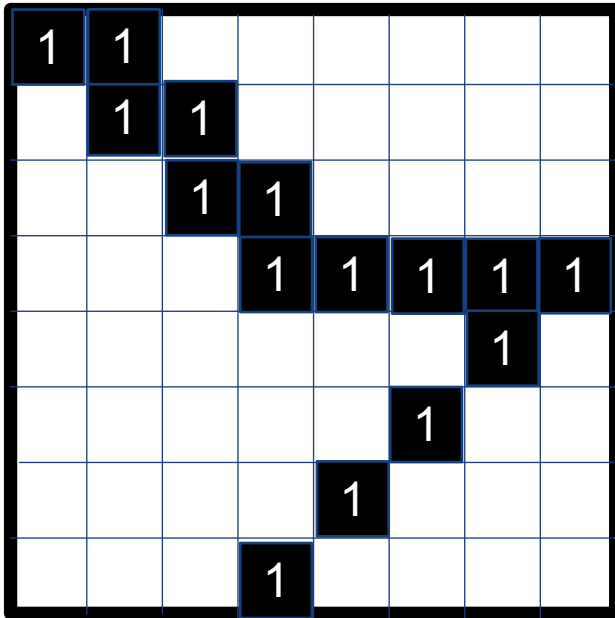
Convolutional neural networks



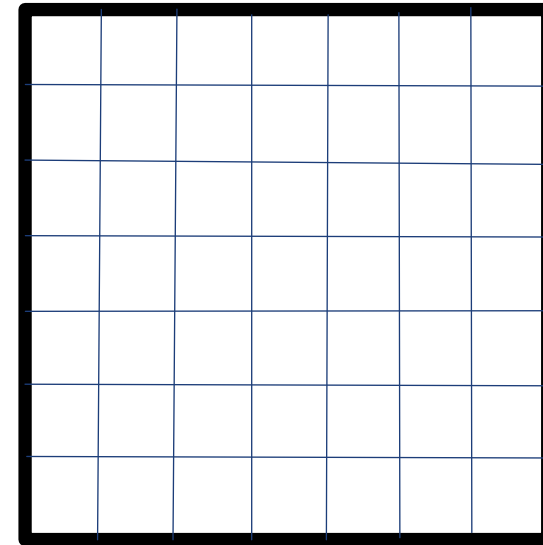
Convolutional filter



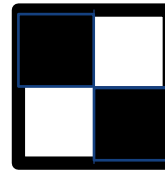
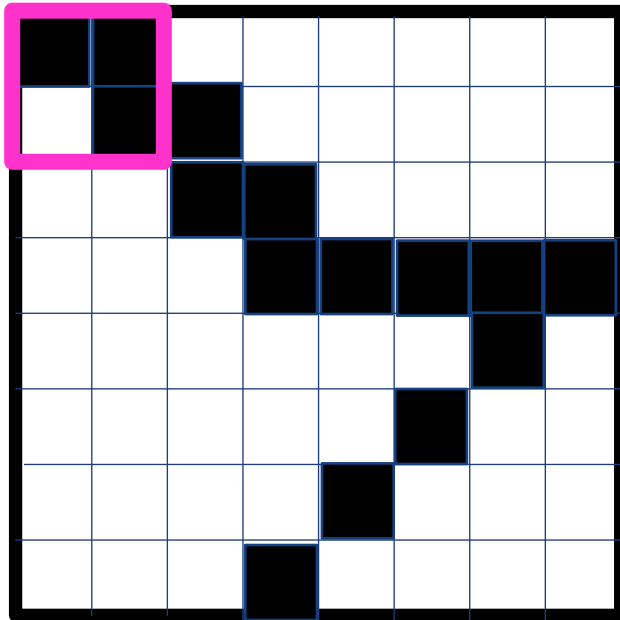
Convolutional neural networks



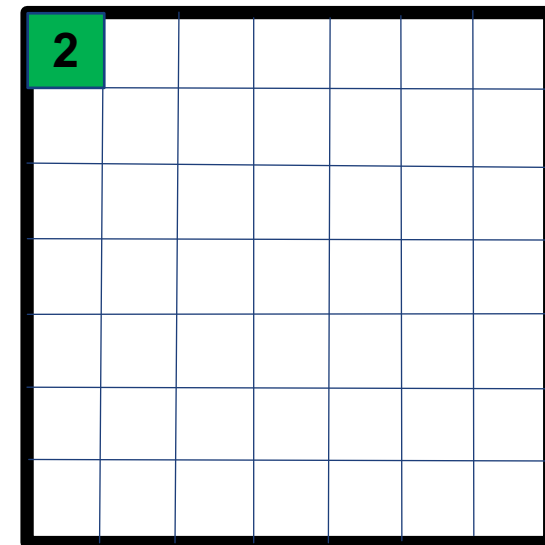
Convolutional filter



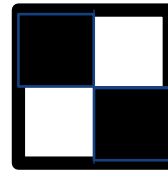
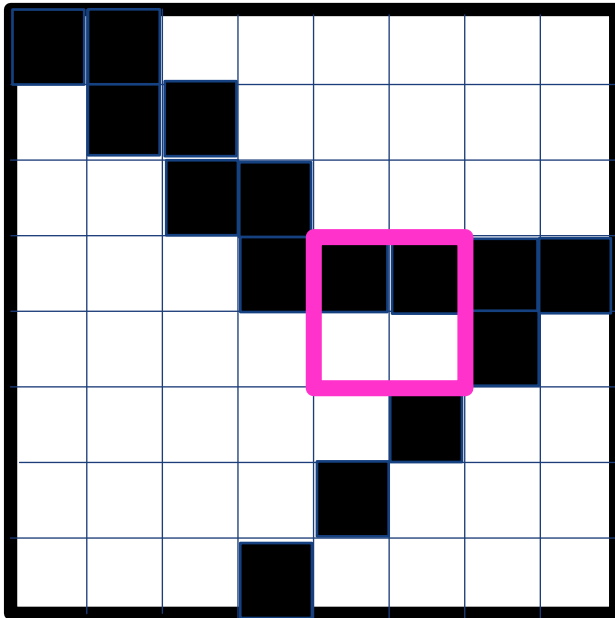
Convolutional neural networks



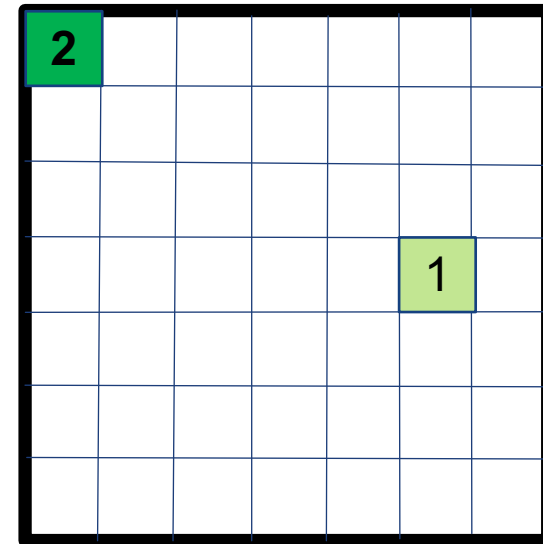
Convolutional filter



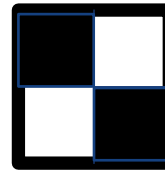
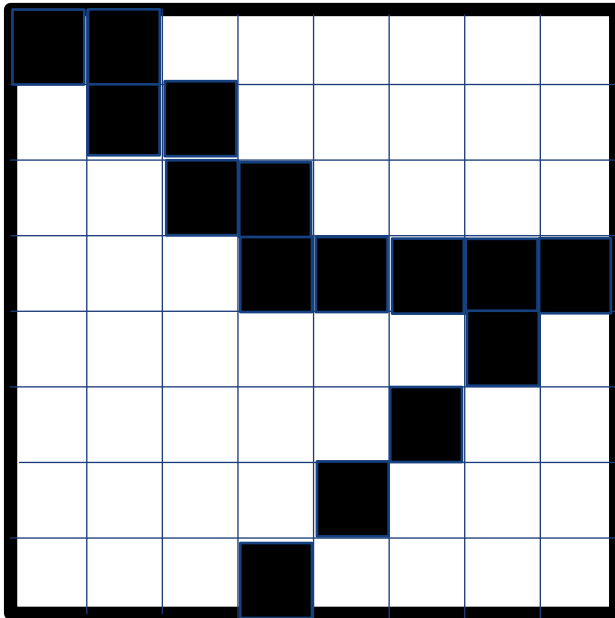
Convolutional neural networks



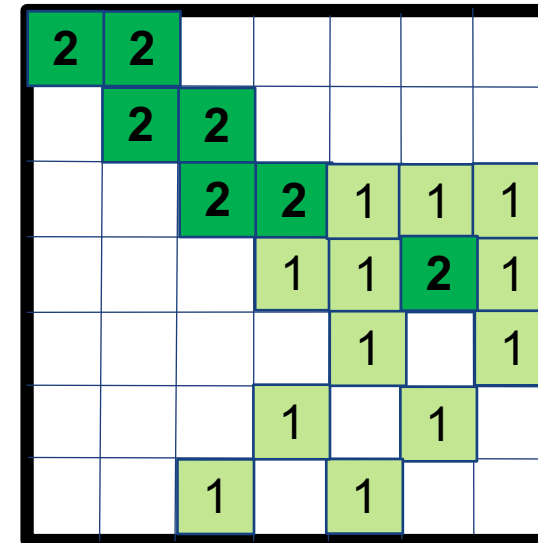
Convolutional filter



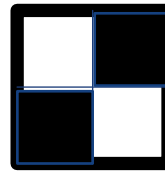
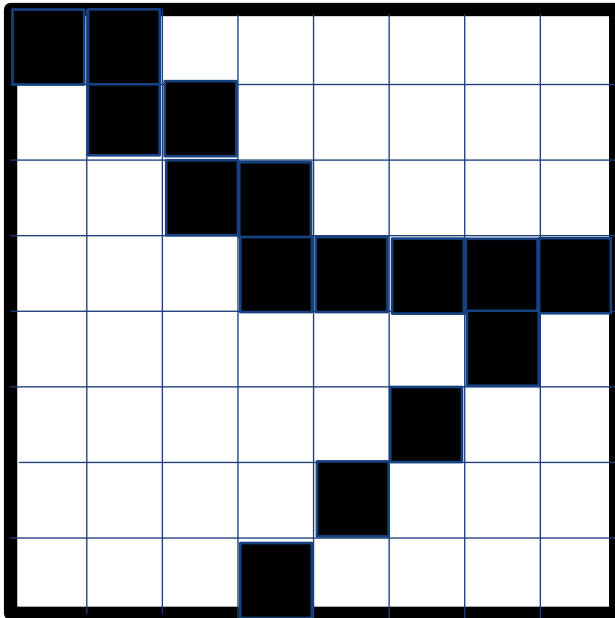
Convolutional neural networks



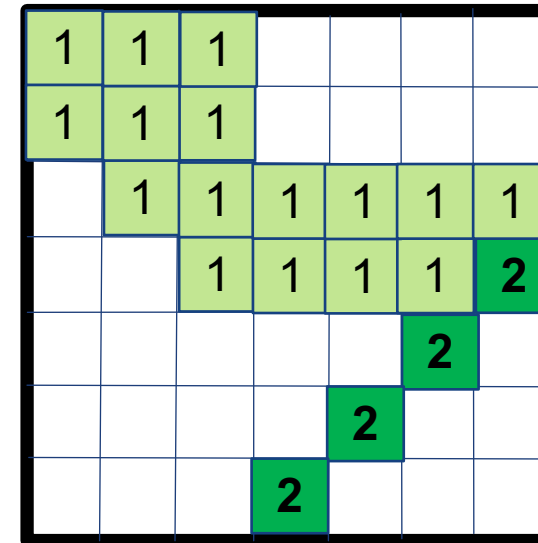
Convolutional filter



Convolutional neural networks



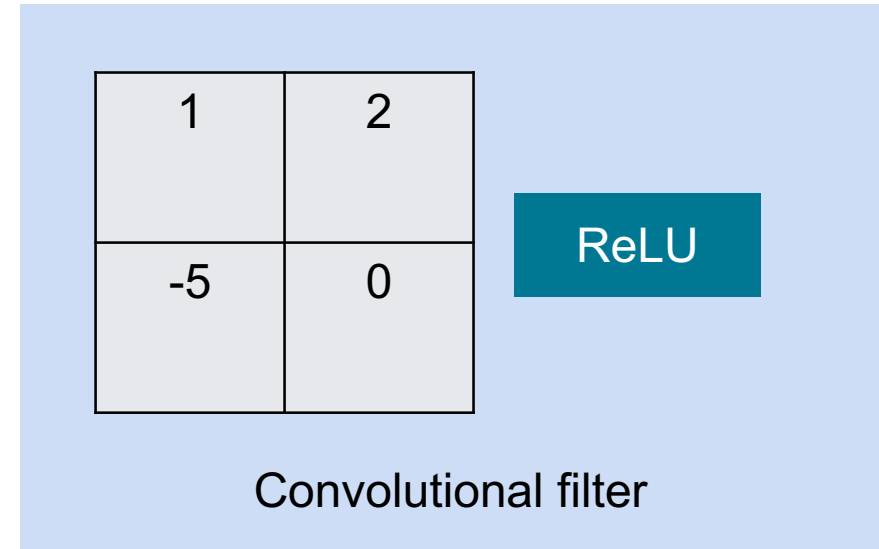
Convolutional filter



Filters

1	3	5
1	2	4
6	3	2

Input

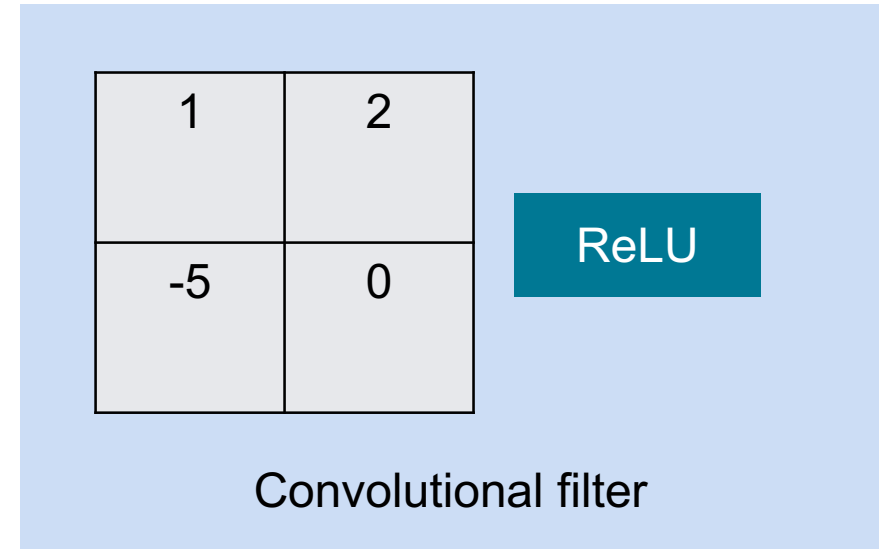


Output

Filters

1	3	5
1	2	4
6	3	2

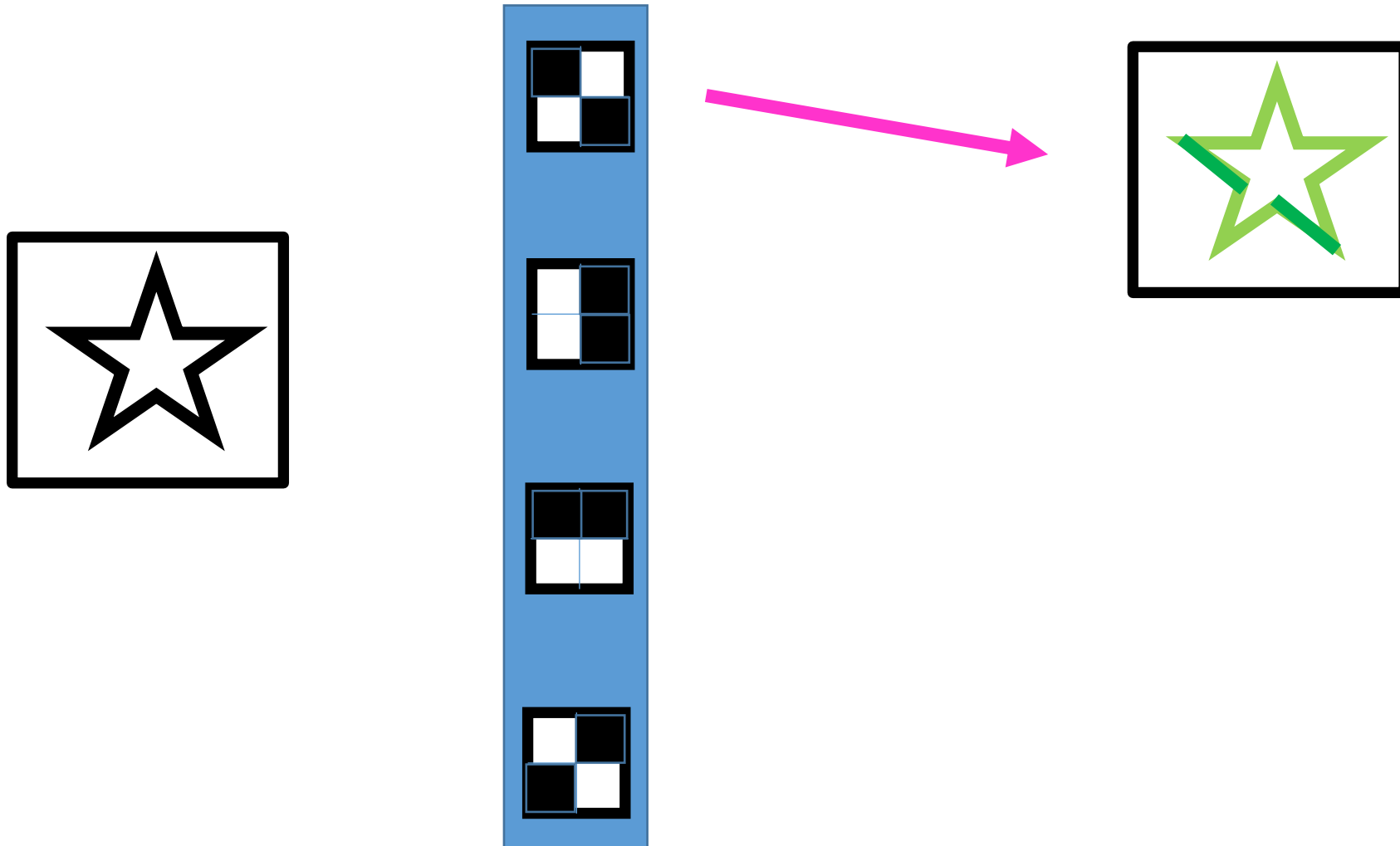
Input



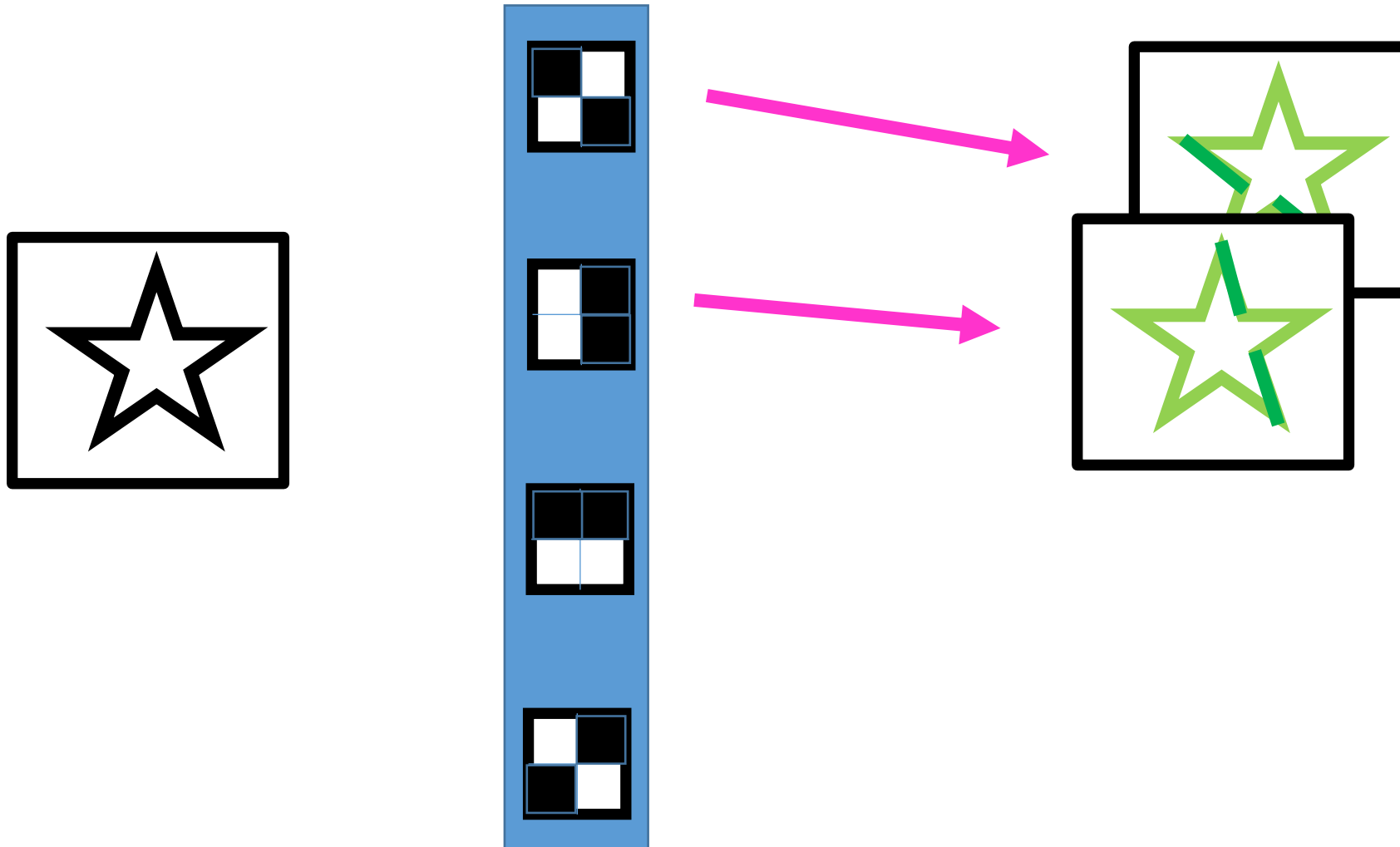
2	3
0	0

Output

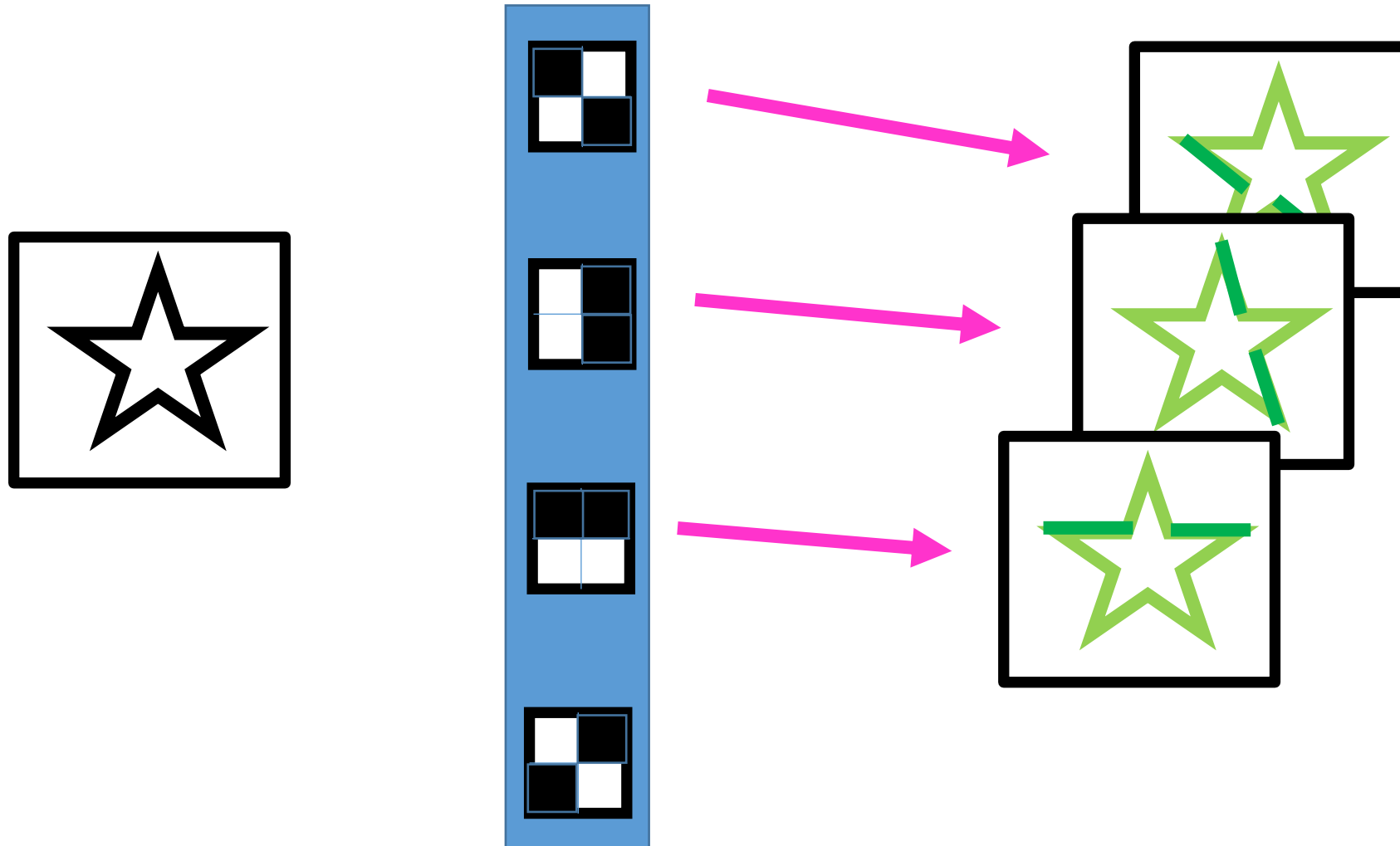
Convolutional neural networks



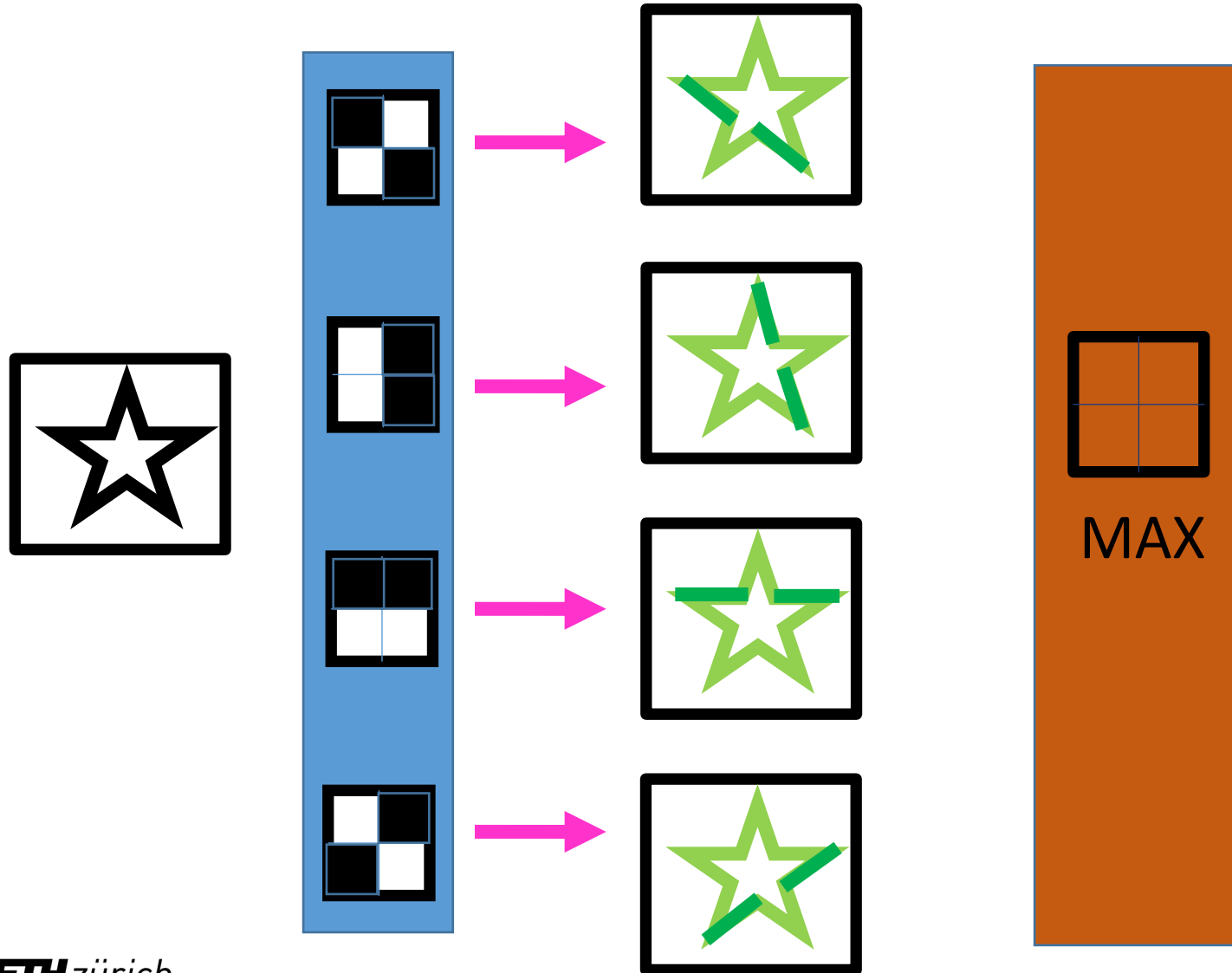
Convolutional neural networks



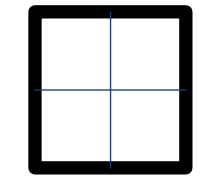
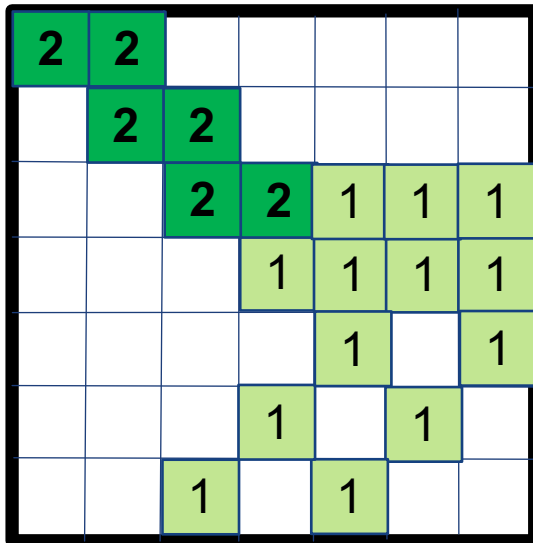
Convolutional neural networks



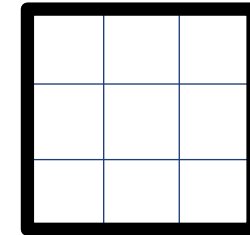
Convolutional neural networks



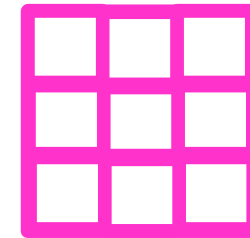
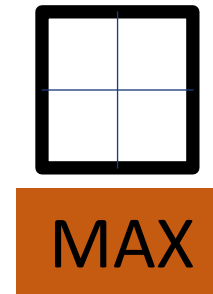
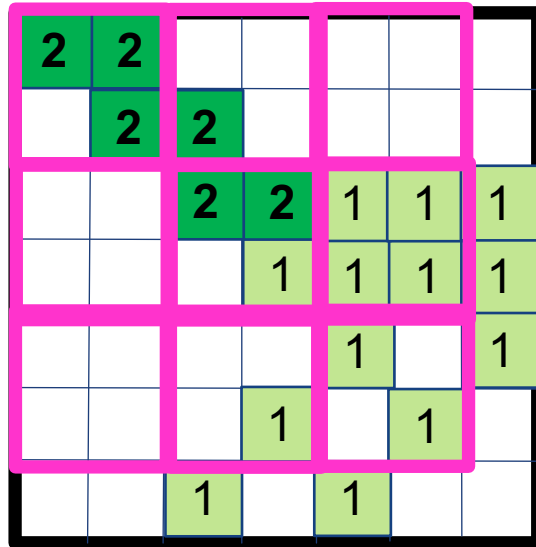
Convolutional neural networks



MAX

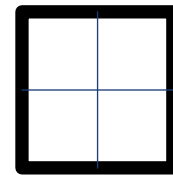


Convolutional neural networks



Convolutional neural networks

2	2					
	2	2				
		2	2	1	1	1
			1	1	1	1
				1		1
			1		1	
		1		1		

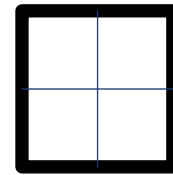


MAX

2	2	
	2	1
1	1	1

Convolutional neural networks

1	1	1				
1	1	1				
	1	1	1	1	1	1
		1	1	1	1	2
					2	
				2		
			2			



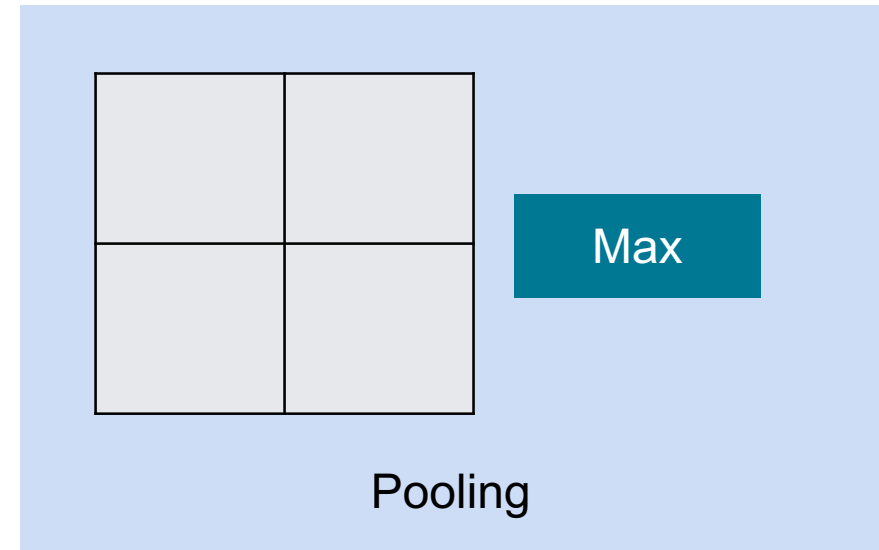
MAX

1	1	
1	1	2
	2	2

Pooling

1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

Input

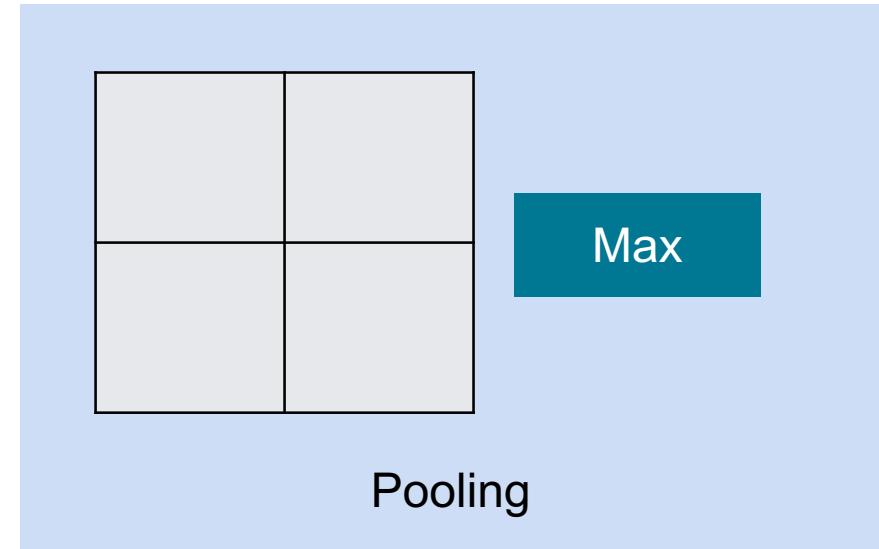


Output

Pooling

1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

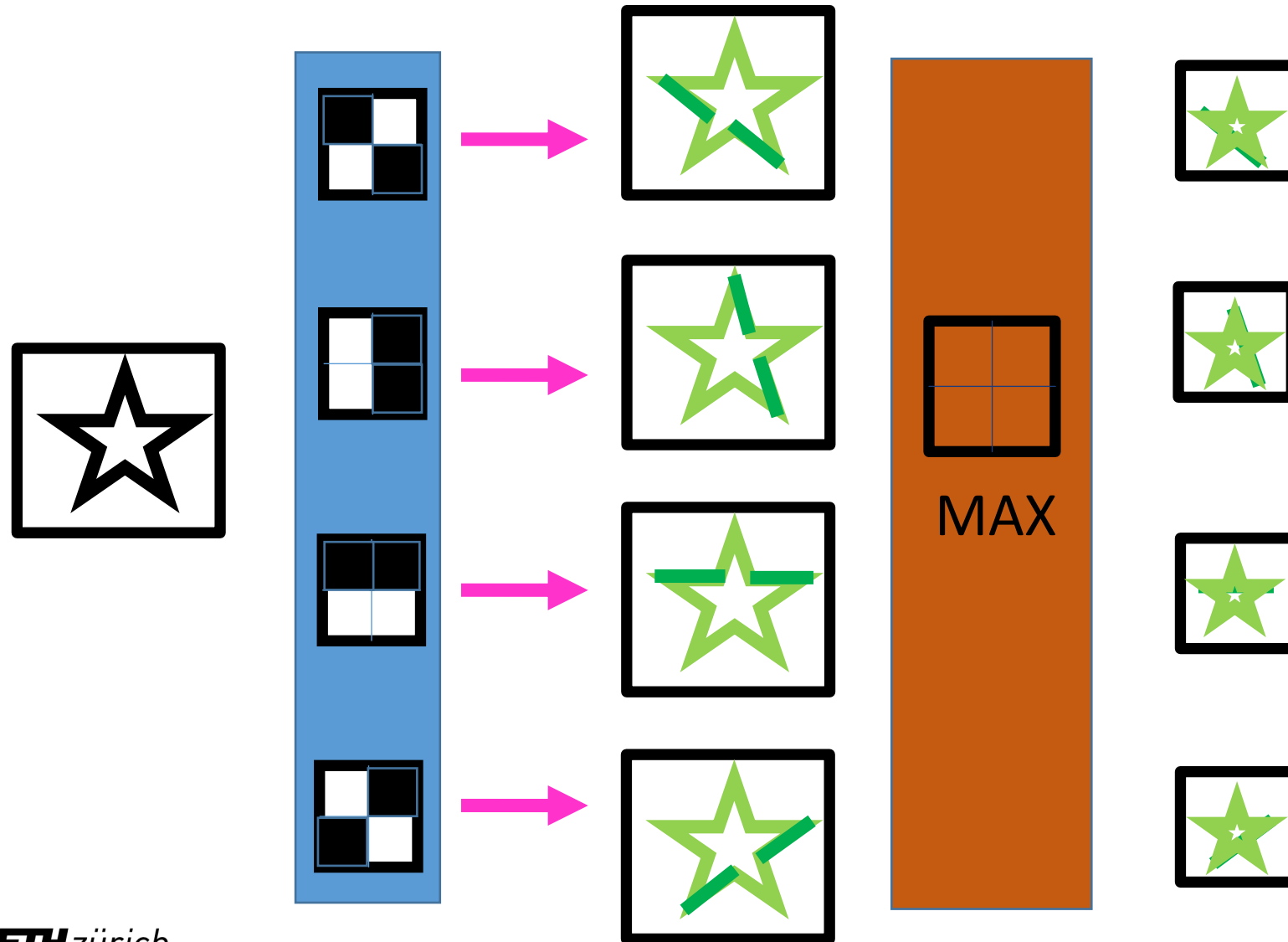
Input



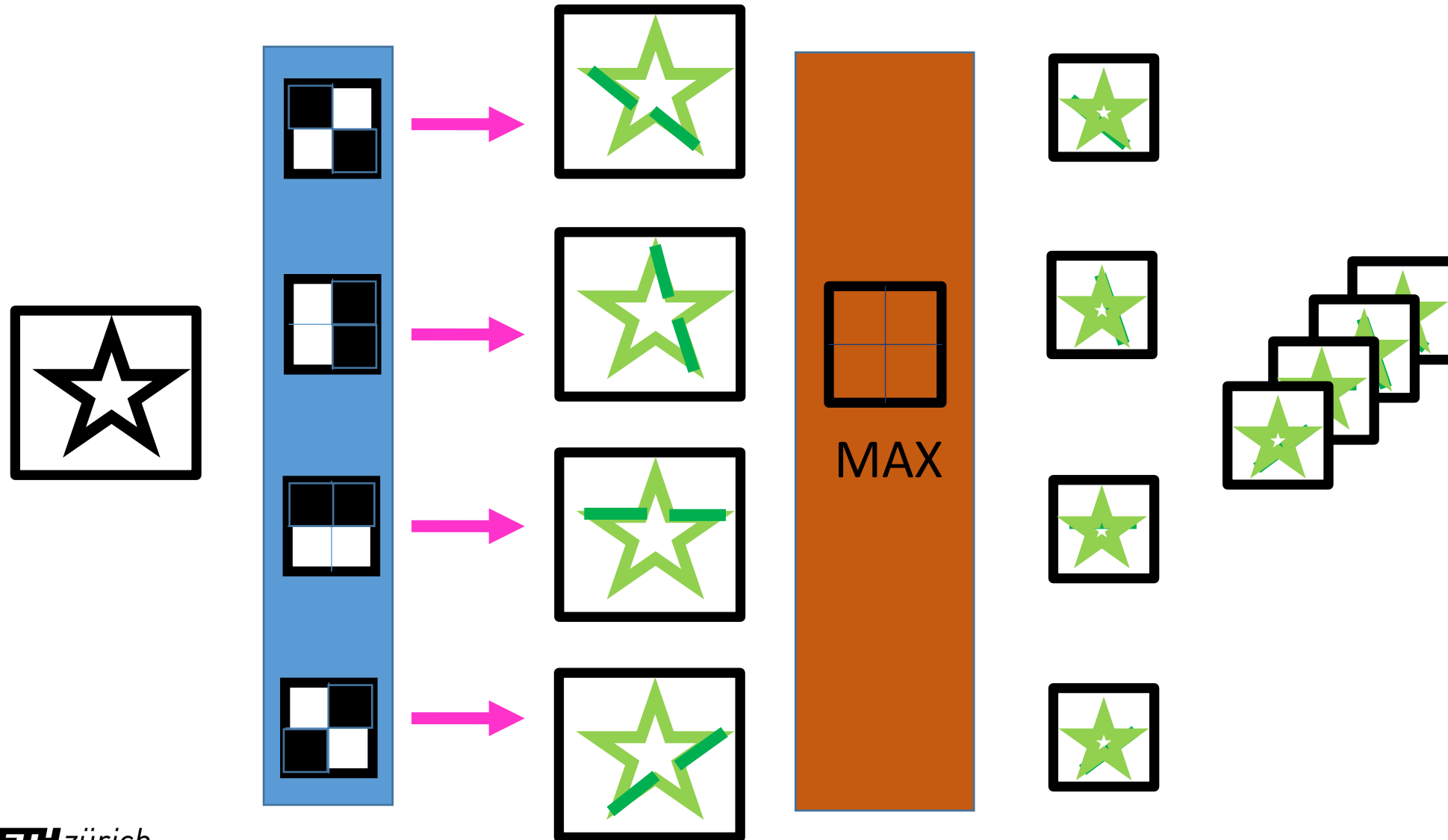
4	5
6	2

Output

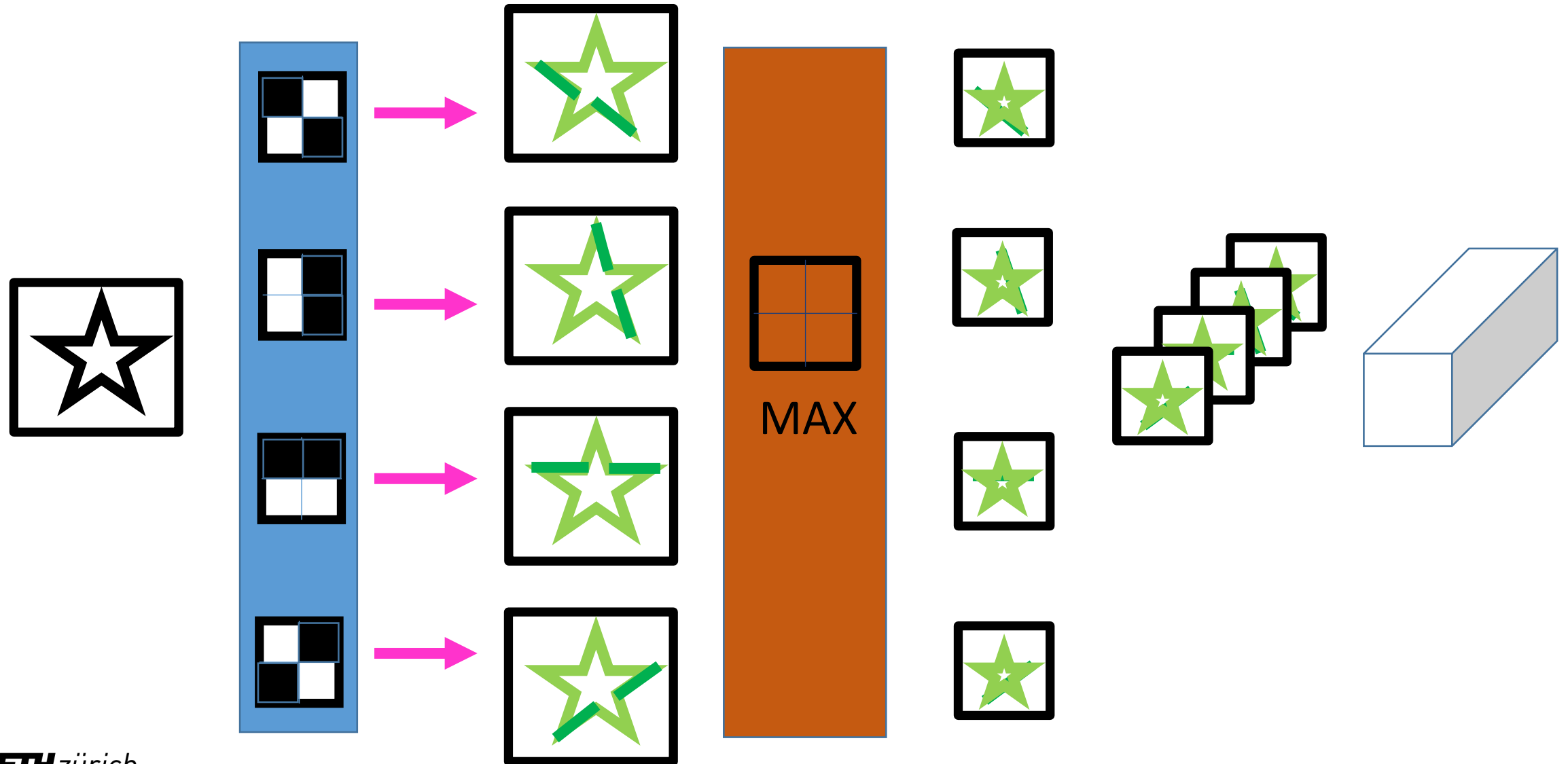
Convolutional neural networks



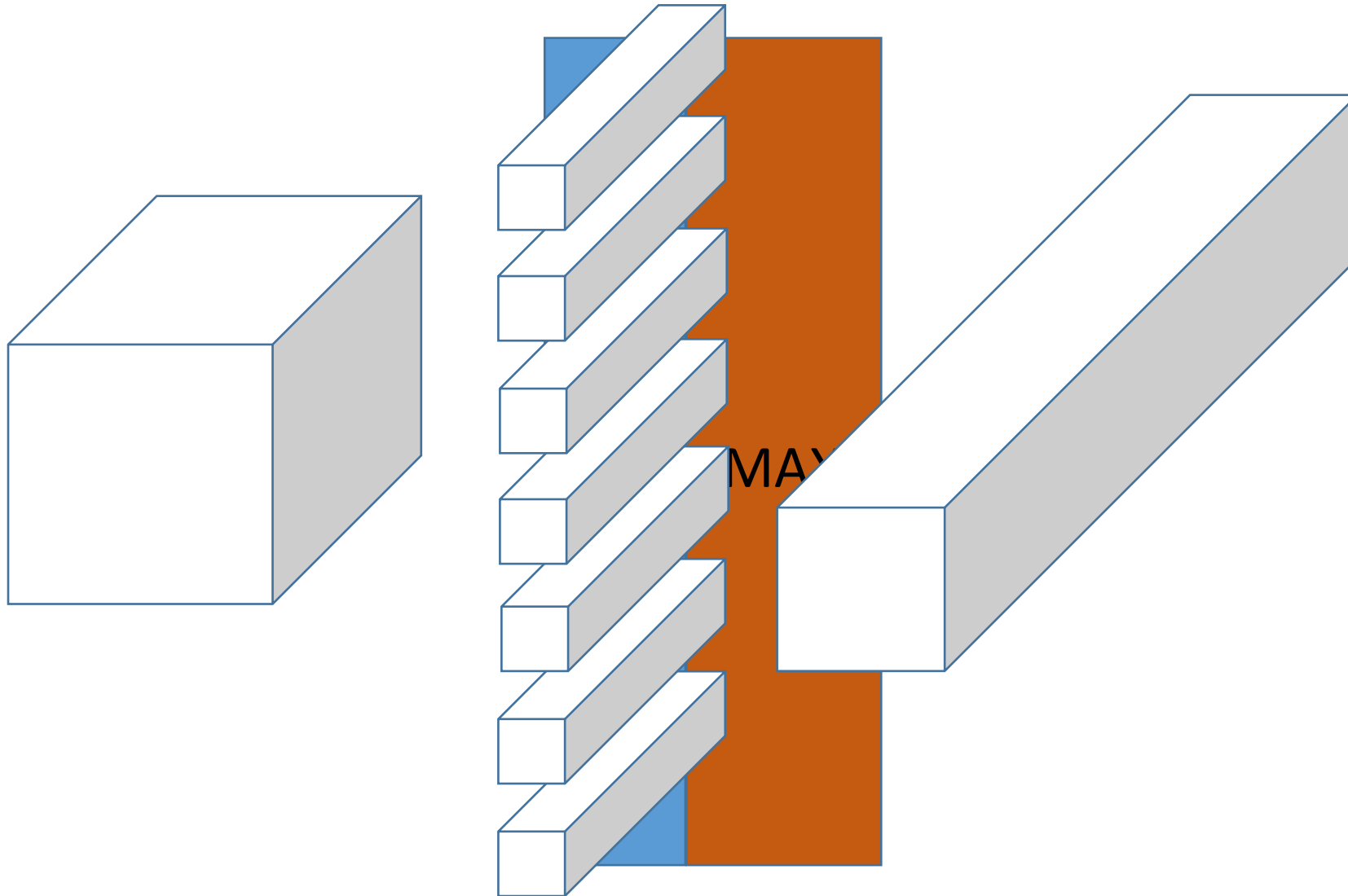
Convolutional neural networks



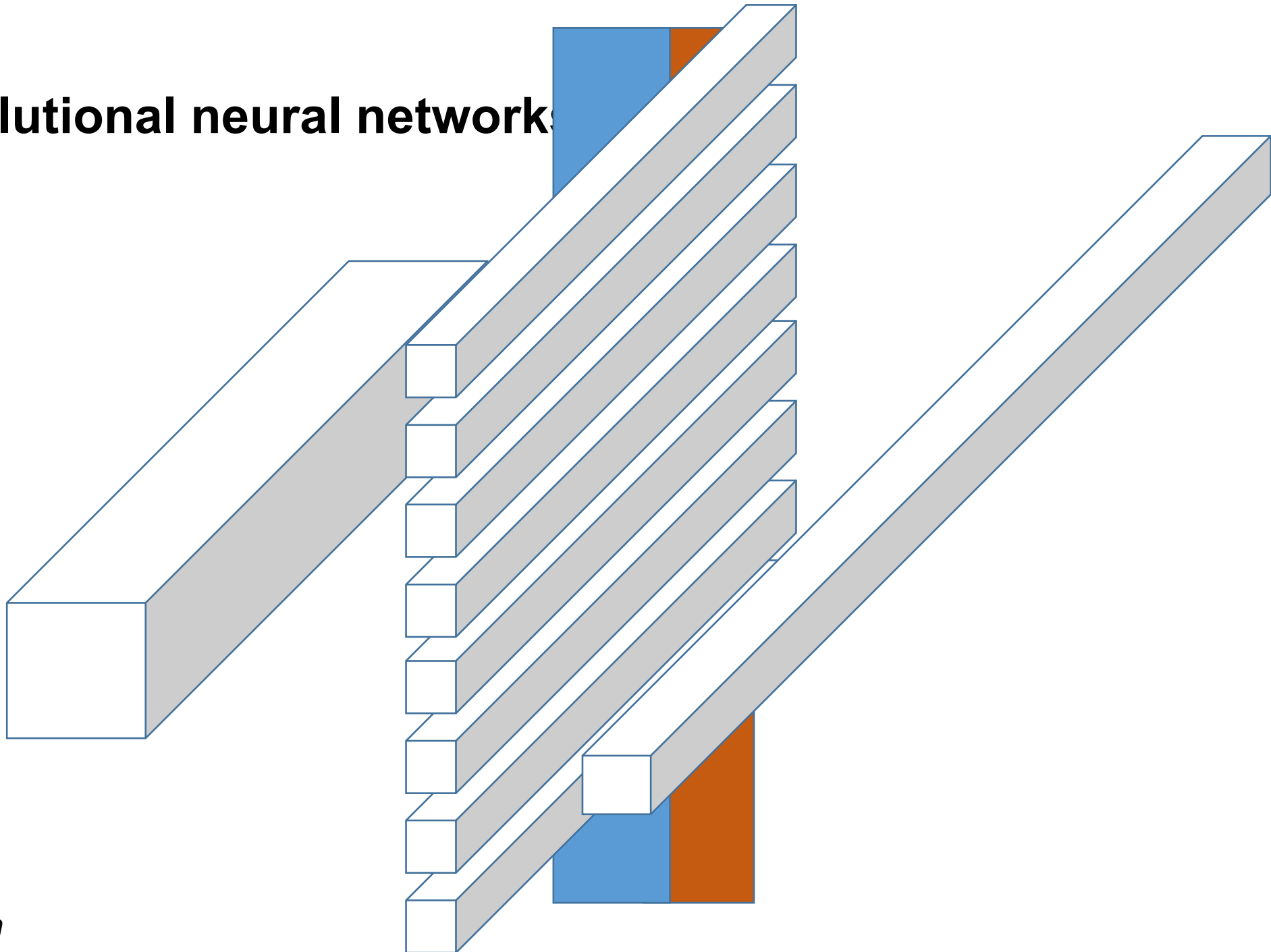
Convolutional neural networks



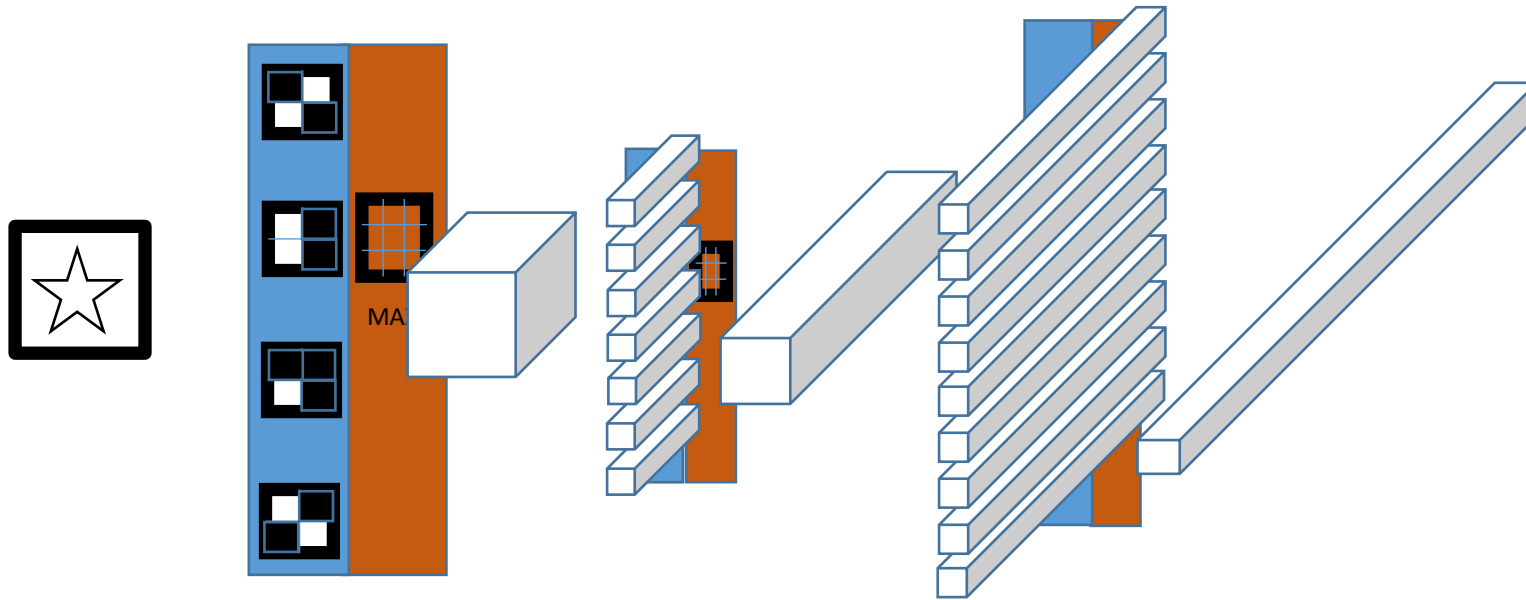
Convolutional neural networks



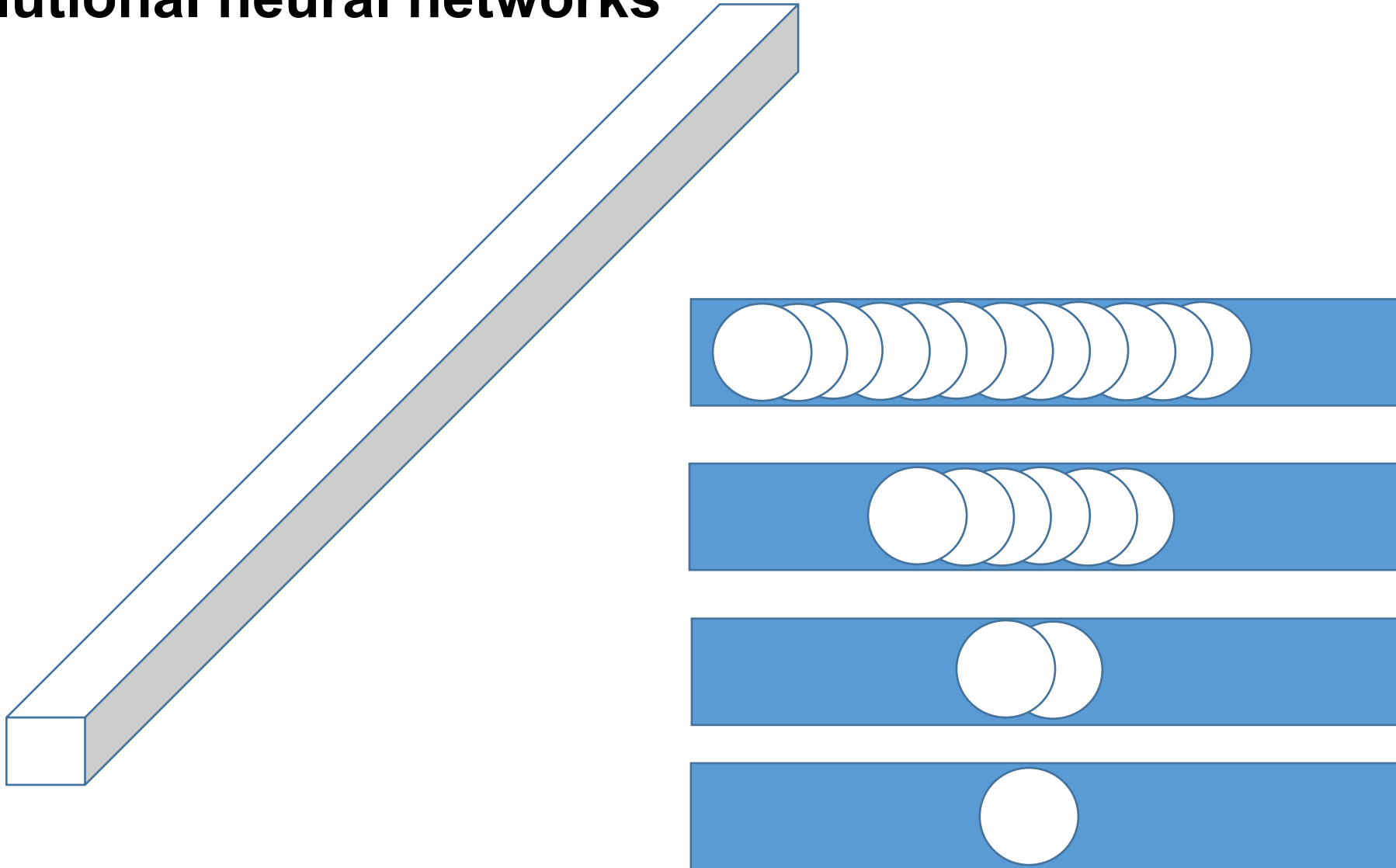
Convolutional neural networks



Convolutional neural networks



Convolutional neural networks



Convolutional neural networks

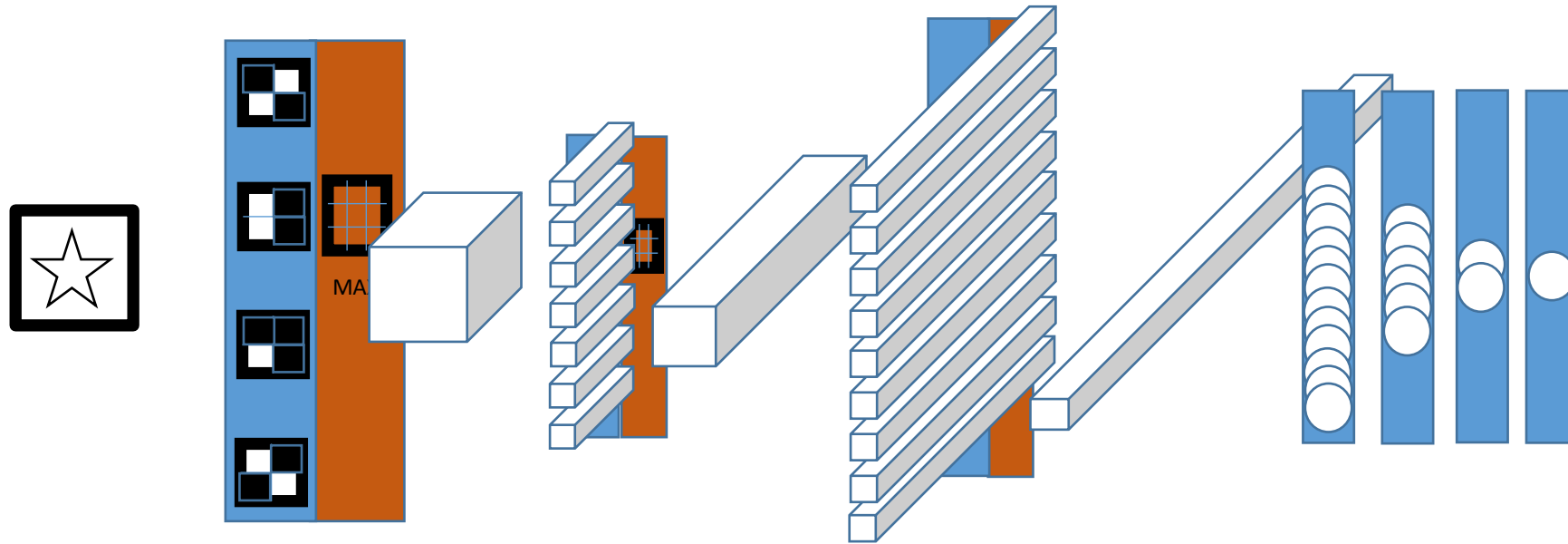
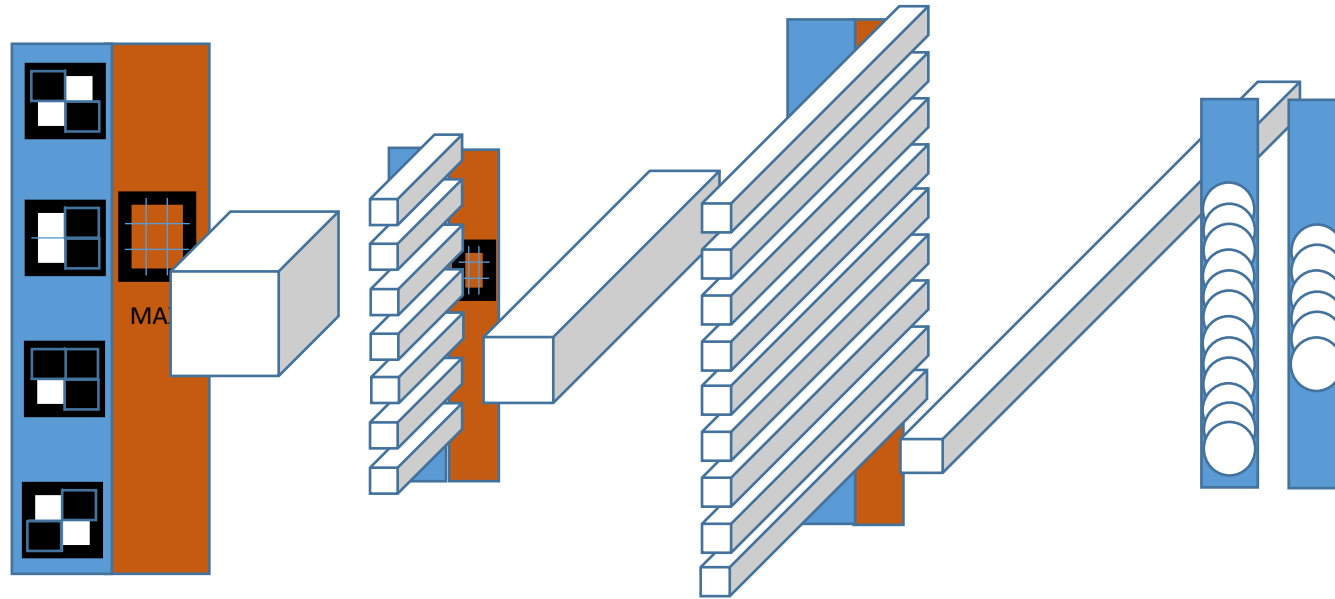
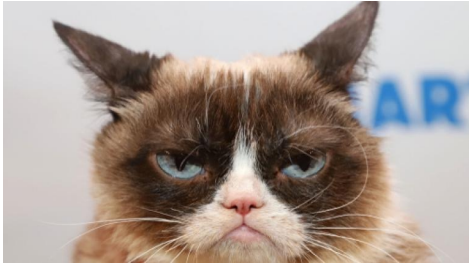
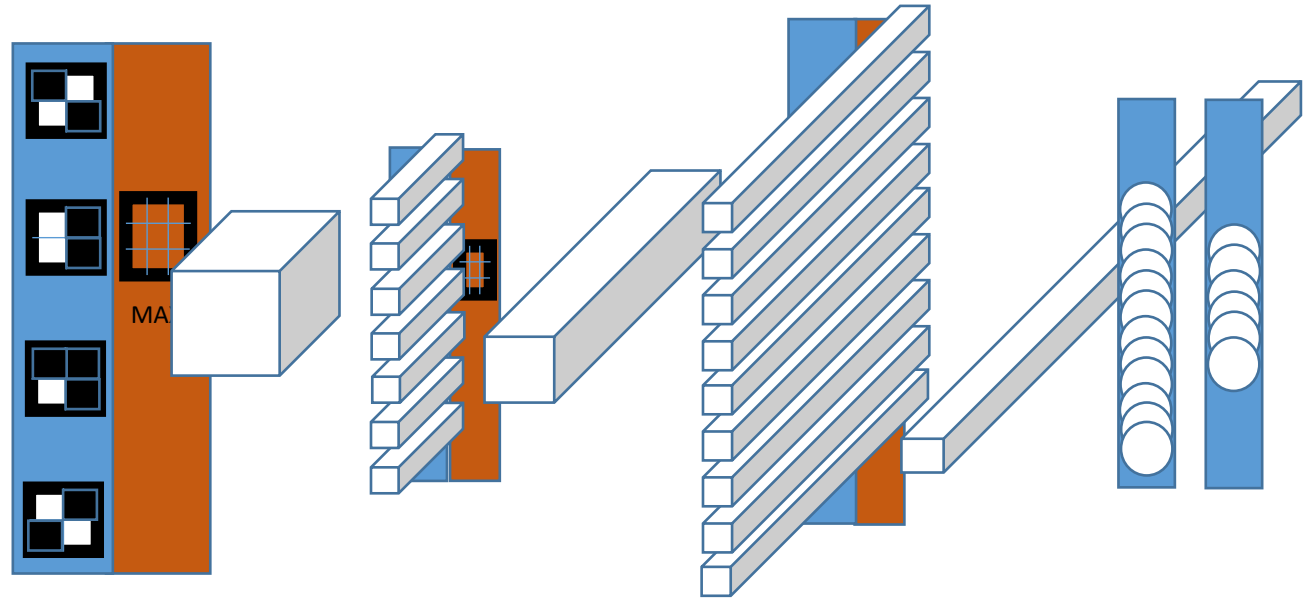


Image recognition



Softmax

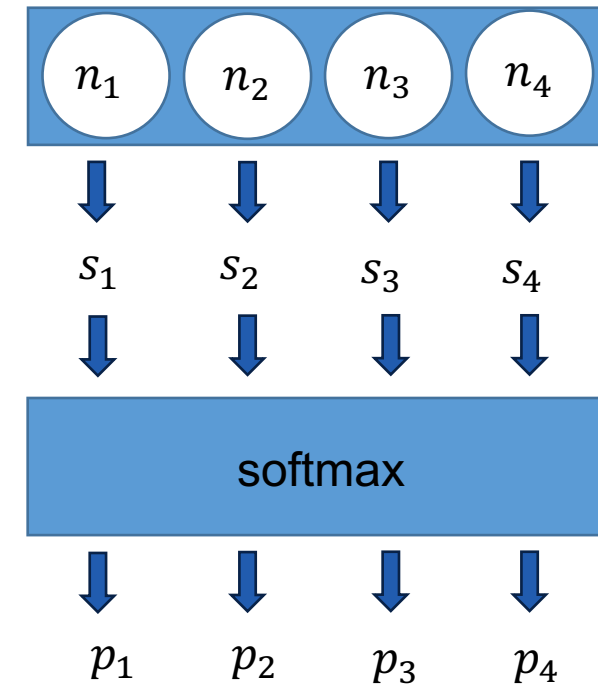


Softmax

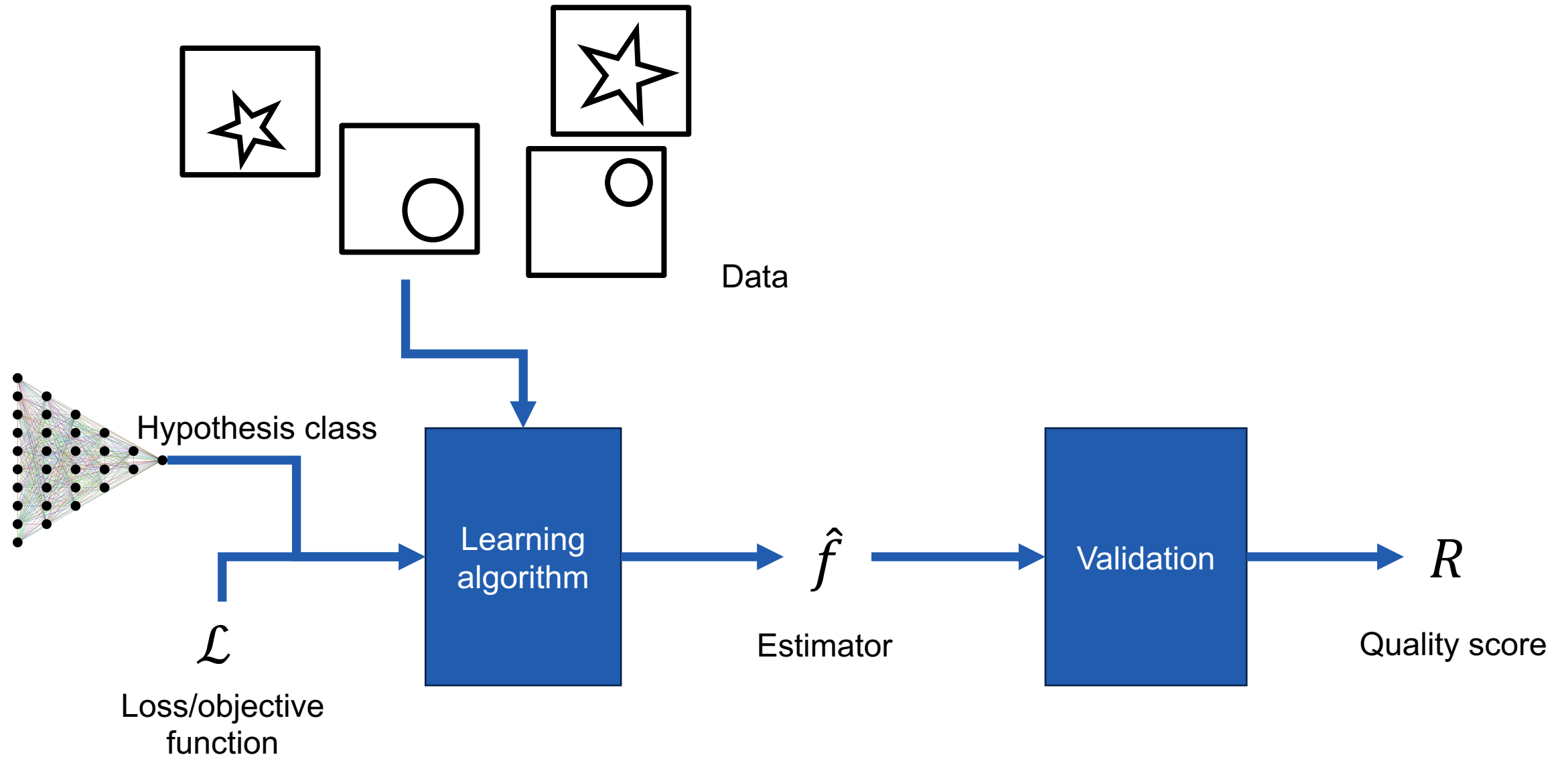


Softmax

- In binary classification tasks, we want to recognize if a particular given object belongs to one of two classes. For these tasks, the CNN has one neuron at the end, whose activation function is the sigmoid function. It produces a value between 0 and 1, indicating the probability that the input belongs to one particular class.
- In multi-class classification tasks, we want to recognize if a particular given object belongs to one of K classes. For these tasks, the CNN has K neurons $n_1, n_2, \dots, n_K$ at the end, each of them producing a score of how much each neuron believes that the object belongs to one of these classes. After these K neurons, we have a softmax function that transforms these values into probabilities $p_1, p_2, \dots, p_K$. Each probability p_i indicates the probability that the object belongs to class i .



Statistical learning in data science

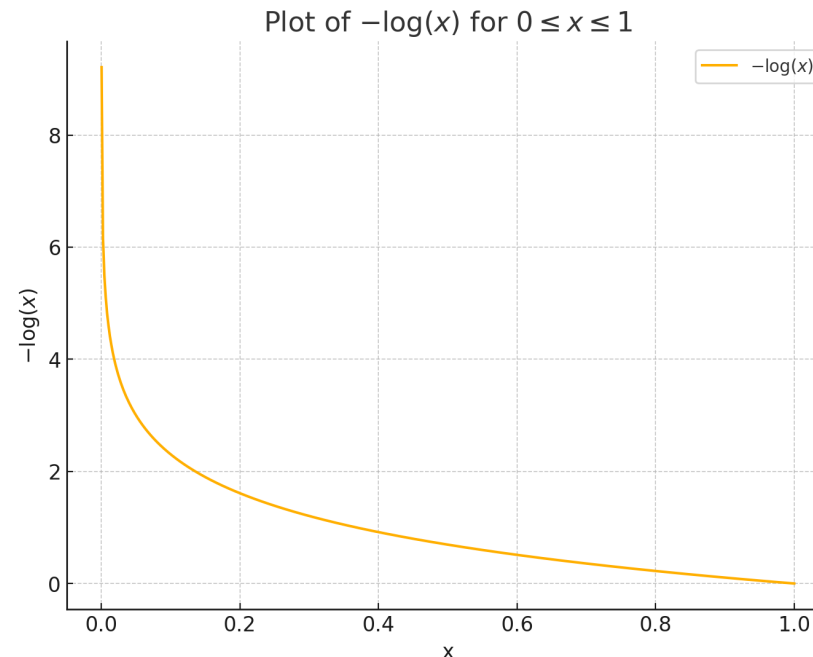


Loss/objective function

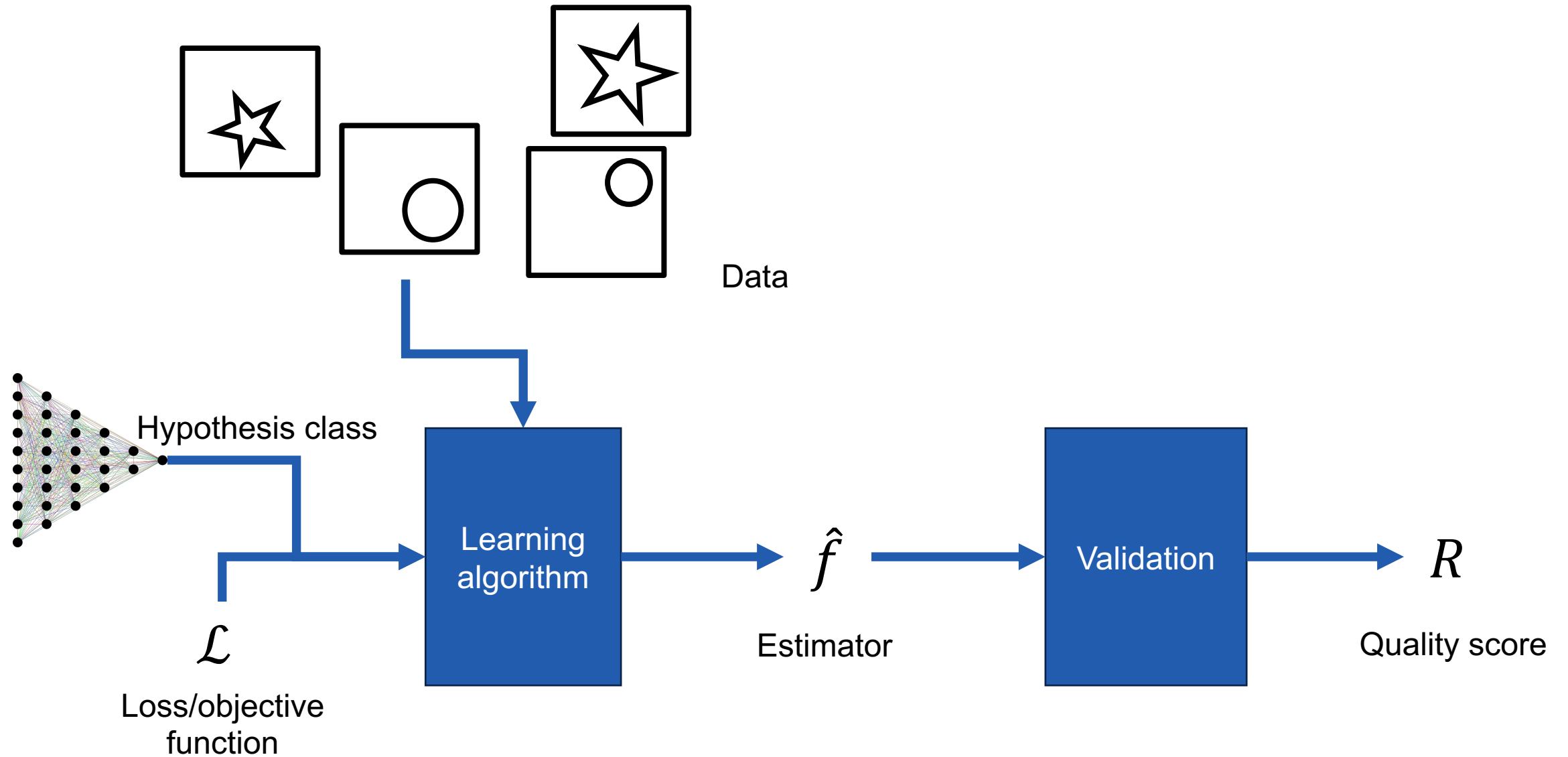
1. A function that takes as input the following:
 - A prediction output by the neural network for an example.
 - The label of that example.
2. The output is a real number that measures how “bad” the prediction is with respect to the example.
3. A popular loss function for classification is the ***cross-entropy loss function***.

Cross-entropy loss function

1. It takes as input the prediction of the neural network for a particular example, represented as a probability distribution $(p_1, \dots, p_K)$ over the K different classes.
2. It takes also as input the actual label $\ell \leq K$ of the example. This label is called **the ground truth**.
3. The output is $-\log p_\ell$.



Statistical learning in data science



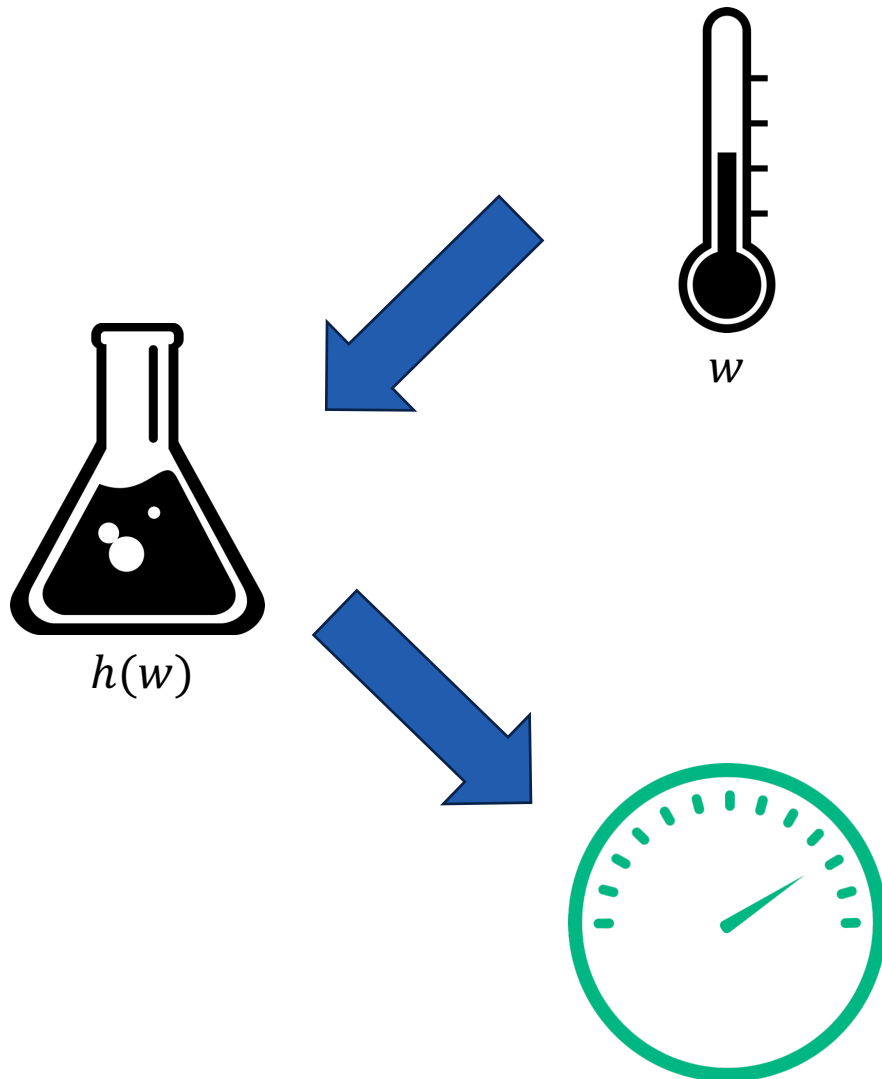
Training: gradient descent



Most ML algorithms use gradient descent to find a good estimator for a given dataset.

1. Make an initial guess for the parameters of your estimator.
2. Evaluate the loss function on your dataset.
3. Compute the gradient of the loss.
4. Correct the parameters of the estimator according to the gradient.
5. Repeat from step 2 until the loss stops decreasing.

Reactor stabilization game



In a cutting-edge chemical research lab, scientists are working on a highly sensitive chemical reaction that needs to be maintained at a precise temperature w^* to ensure the reaction reaches a desired state. This state corresponds to the "root" of a function h related to the reaction's equilibrium.

However, we do not have access to the function h directly. We get the pH level through a noisy device that approximately measures the pH level in the reaction.

Link to the game:

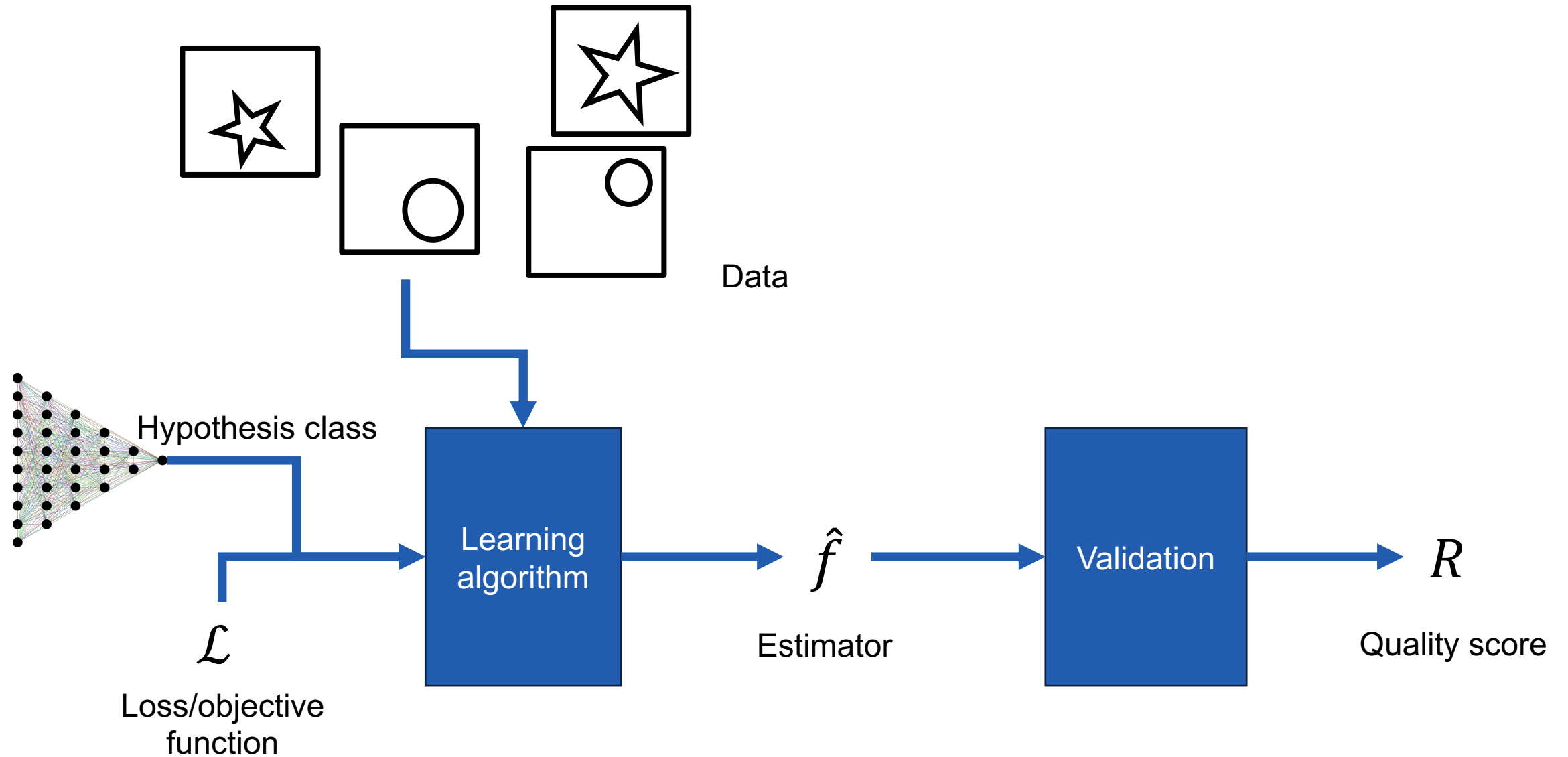
https://colab.research.google.com/drive/1HC-o-C4bSCPUscAiX7BiGcQOdM_FO-Zym?usp=sharing

$$\Delta(w) = h(w) + \epsilon, \text{ with } \mathbb{E}[\epsilon] = 0.$$

Connection to gradient descent

- We learned to reach the pH level of 7 by noting that lowering the temperature lowers the pH level and increasing the temperature increases the pH level.
- When we have an analytical function $\mathcal{L} : \mathbb{R}^d \rightarrow \mathbb{R}$, then we can compute its gradient $\nabla \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \theta}$. The gradient points in the direction that decreases $\mathcal{L}$ the fastest.
- The gradient shows how to tweak the parameters of a neural network in the way that decreases the loss function the fastest.

Statistical learning in data science



Validation

- Once you have computed an estimator (i.e., a neural network) using gradient descent, you must validate it on a separate dataset!
- Why? Too complex neural networks may risk overfitting your dataset!